Do LLMs Brainstorm Better with Knowledge Graphs?

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Abstract

In AI-assisted product ideation, describing the problem as a knowledge graph is an attractive way to represent the solution space, and one that seems well suited to LLM-powered combinatorial innovation and search. Yet most studies benchmark the output on innovation metrics; few ask how the form of the graph given to the model affects what it produces, and at what cost. We test this on a microwave oven described as a graph (123 nodes, 246 edges) linking user needs to the functions, parts and technologies that serve them. For each of 13 needs, `claude-haiku-5-5` got either (i) the whole graph, (ii) only the slice relevant to that need, (iii) a random slice of the same size, or (iv) no graph, at four levels of thinking effort. `claude-sonnet-5-5` then scored the resulting 2,892 ideas from 580 runs blind, and we counted top ideas (roughly the best 12%) per 100k tokens. A follow-up asked three models (`claude-haiku-5-5`, `claude-opus-5-5` and `claude-fable-5-1`) for 50 ideas per call on one fixed market need, at a fixed (medium) effort, adding 6,091 more ideas.

Attaching the full graph to the prompt never paid off. In the first study it yielded about 3 top ideas per 100k tokens, against 8.8 for the relevant slice and 6.4 for no graph. In the follow-up it was the least efficient condition for every model, at about half the yield of the slice or no graph for Haiku and Fable. For the stronger models the full graph did not lower the share of good ideas; it only added cost. Among the ways of giving the graph, the relevant slice worked best, but within the follow-up's uncertainty it was no better than giving no graph at all. The stronger models found two to three times as many top ideas per token, yet at list prices Haiku found 20 to 40 times more per dollar, an advantage that holds when an automated judge screens the ideas but reverses when people read them. Thinking longer raised quality but not efficiency. An expert who read every idea of the first study found that even the best mostly restate existing technology: the model retrieves the state of the art more than it invents.

1 Introduction

Most research on AI for innovation asks about outcomes: whether LLM-generated ideas rival human ones (Si et al., 2024; Terwiesch et al., 2026), whether AI assistance elevates individual or team creativity (Doshi and Hauser, 2024; Anderson et al., 2024), and whether agentic pipelines outperform single prompts or human designers (Liu et al., 2026; Massoudi and Fuge, 2026). Yet outcome benchmarks against human baselines expire rapidly as model capabilities expand

*Code, data, evaluation traces, and interactive visualizers: `https://github.com/orioli/IM-LLM`

and token prices fall. A more enduring question is how to **present the problem** to the model: which representation of a product allows generative AI to turn compute into standout innovation concepts most efficiently.

In product ideation, engineering design, and service innovation, problem representations range from business model canvases and user journey maps to design structure matrices and typed knowledge graphs (Akao, 2024; Steward, 1981; Gero, 1990). Here, we focus on the *innovation matrix* (IM) of Leo Tchirsky (Berengueres, 2023)—a layered, typed graph connecting market needs to functions, components, enabling technologies, and scientific fields. While cost-aware evaluation is now standard for LLM reasoning and agent benchmarks (Chen et al., 2024a; Wang et al., 2024; Kapoor et al., 2025), generative ideation is still largely judged by average scores without accounting for token expenditure or the extreme-value nature of brainstorming (Girotra et al., 2010; Dahan and Mendelson, 2001).

To address this gap, we examine how problem representations and reasoning allocations govern AI brainstorming efficiency across four questions: which graph representation yields the most top ideas per token (RQ1); whether structural relevance or raw prompt length drives performance (RQ2); how investing tokens in context compares with spending them on thinking effort (RQ3); and whether the effect of the representation holds across models, and what each model costs per top idea (RQ4). We test these questions on a countertop microwave oven represented as an innovation matrix with 123 nodes and 246 edges, using the *Substitute* operator from the SCAMPER ideation framework (Eberle, 1971; Osborn, 1953). We compare four prompt representations: the full graph (A), a targeted vertical slice of the active market need (functions, components, technologies, fields, linked constraints and actors; B), a size-matched random slice (R), and a no-graph control (C). In Experiment 1 (RQ1–RQ3), `claude-haiku-5-5` receives each representation for 13 market needs at four thinking-effort levels, in 580 API calls generating 2,892 ideas. In Experiment 2 (RQ4), three models, `claude-haiku-5-5`, `claude-opus-5-5` and `claude-fable-5-1`, each receive the four representations for one market need and are asked for 50 ideas per call, in 120 calls generating 6,091 ideas. All ideas are scored blind by `claude-sonnet-5-5`, and Experiment 1's ideas were also read by an expert. We measure *yield*: top ideas (score $\geq$ 4.0/5.0; the top $\sim$12% in Experiment 1) per 100k tokens spent, and in Experiment 2 also per US dollar at list price.

Our primary finding is that prompting with a targeted market-need slice at medium effort achieves peak yield (8.8 top ideas per 100k tokens), nearly tripling the efficiency of the full graph. Note here that the size of the slice is about one third that of the full graph. Although the full graph produces more specific average sets, it yields no more top-tail ideas while doubling prompt costs. Furthermore, scaling thinking effort raises average linguistic quality but fails to improve token efficiency. Together, these findings make three contributions: they reframe AI ideation around representation efficiency (outcome per token) rather than ephemeral human baselines; they establish yield as a non-saturating tail metric that captures extreme-value creative discovery where mean scores mislead; and they provide an open, controlled empirical comparison of product-graph representations and thinking effort, with all prompts, calls, and judgments released.

The remainder of this paper is structured as follows. Section 2 reviews related work on design representations, LLM ideation workflows, and cost-aware evaluation. Section 3 details the product graph, prompt representations, experimental design, and scoring metrics. Section 4 presents the findings for RQ1–RQ3 (Experiment 1) and RQ4 (Experiment 2), and robustness checks. Finally, Section 5 discusses implications for design thinking and outlines directions for graph-guided AI innovation.

2 Related work

2.1 Innovation Matrix background

Structured descriptions that link what customers want to what a product does and what it is made of are older than computing in design. Quality function deployment began in Japanese manufacturing in the late 1960s, was taken up by Toyota and its suppliers in the 1970s, and reached Western industry as the "house of quality" (Akao, 2024; Hauser and Clausing, 1988; Chan and Wu, 2002). Its matrices relate customer requirements to engineering characteristics, and then to parts and processes, so that every design decision can be traced to a market need. Related schemes followed, each making a different link explicit: morphological analysis lays out the alternatives for each function (Zwicky, 1967; Ritchey, 2006, 2018); the design structure matrix records the dependencies between components (Steward, 1981; Browning, 2001; Eppinger and Browning, 2012); function–behaviour–structure separates what a design is for from what it is (Gero, 1990; Gero and Kannengiesser, 2004); the functional basis standardises the vocabulary of functions and flows (Stone and Wood, 2000; Hirtz et al., 2002); and technology roadmaps link market needs to products and technologies over time (Phaal et al., 2004). The innovation matrix used here (Berengueres, 2023) combines several of these links in one layered graph, from market needs through functions and components to technologies and scientific fields (it is distinct from the design-led innovation matrix of Kyffin and Gardien (2009)). All of these were made for people, typically a team in front of a wall chart, who read them selectively.

2.2 LLMs that innovate with design methods

Language models were first used as concept generators (Zhu and Luo, 2022; Ma et al., 2023) and compared with people: their concepts are less diverse than crowdsourced ones (Ma et al., 2025), their research ideas are judged more novel but slightly less feasible than experts' (Si et al., 2024), and the effects of working with them are mixed: stories became more creative but more similar to each other (Doshi and Hauser, 2024), and groups less semantically diverse (Anderson et al., 2024), whereas high exposure raised collective idea diversity without changing the creativity of individual ideas (Ashkinaze et al., 2025), and an image generator led to more design fixation and fewer, less varied and less original ideas (Wadinambiarachchi et al., 2024). Large language models are often conceptualized as a lossy, blurry copy of the web (Chiang, 2023), distilling vast online corpora into compact neural weights (Karpathy, 2023). Recent systems give the model the structured methods above. An FBS ontology steers concept generation (Chen et al., 2024e); generated text and images enrich concepts (Chen et al., 2024b); TRIZ is applied by single models and by agent teams (Chen et al., 2024c; Jiang et al., 2025; Szczepanik and Chudziak, 2025); morphological analysis is automated (Chen et al., 2024d); and agentic workflows chain whole design processes (Ding et al., 2023; Massoudi and Fuge, 2026). iDesignGPT (Liu et al., 2026) goes furthest: agent clusters run problem definition with QFD and Kano analysis, then functional decomposition, morphological analysis, SCAMPER (Eberle, 1971) and TRIZ, and produce a product design specification.

2.3 What "innovative" is checked against

These systems claim to innovate, but the claim usually rests on ratings: experts or a model score novelty and originality, or embedding distances measure how far the generated concepts spread from each other. iDesignGPT, for example, reports "competitive novelty and consistently higher originality and modularity" than GPT-4o zero-shot, GPT-4o chain-of-thought and Deepseek-r1, on the basis of expert scores and semantic novelty metrics; patents are searched as an input to the workflow, and originality is scored as the embedding similarity of the generated designs to patents (lower is more original), but the score ranks designs against each other and does not establish whether an idea already exists (Liu et al., 2026). This is typical. Novelty is judged

relative to other generated ideas or to a rater's memory, seldom against what already exists, although patent-based measures of technological novelty (Fleming, 2001; Verhoeven et al., 2016) and citation-based measures of novel combinations in the literature (Uzzi et al., 2013) are well established, and patent-derived knowledge networks are available for design (Sarica et al., 2020; Luo et al., 2021). An idea a rater finds novel may be on the market already. Our own study shares this limitation: its judge rates novelty from knowledge, without a prior-art search, and several of its top-rated ideas, such as solid-state RF heating and RF thawing, already exist in industry, as the judge itself noted. A second gap is cost: few of these systems report what their outputs cost, and none compares configurations at equal cost, so a more elaborate pipeline cannot be weighed against a simpler one. To our knowledge, none of these studies reports a patent resulting from its generated concepts. Nor do the workflows described include collecting first-hand information in the field (*genchi genbutsu*, "go and see" (Berengueres, 2023)).

2.4 Context and cost in LLMs

More context is not free even when it fits in the window. In practice, models exhibit an "effective context window" far smaller than their advertised capacity, with performance degrading sharply as input token length increases (Paulsen, 2026), a limitation attributed to the left-skewed frequency of relative positions in training, which leaves long-distance positions under-trained (An et al., 2024). Models use information in the middle of long inputs poorly (Liu et al., 2024), their reasoning degrades as inputs grow long well before the limit (Levy et al., 2024), and they struggle when relevant content must be inferred rather than matched (Modarressi et al., 2025); retrieval is much cheaper than long context, which still performs better when resourced sufficiently, and routing between the two reaches comparable performance at lower cost (Li et al., 2024). Elsewhere, cost is becoming a first-class outcome: cascades and routers match strong models at lower cost (Chen et al., 2024a; Ong et al., 2024; Ding et al., 2024), reasoning strategies are compared at equal token budgets (Wang et al., 2024; Han et al., 2025), and agent benchmarks report accuracy together with cost (Kapoor et al., 2025; Wang et al., 2025). This concerns tasks with defined right answers. Ideation, as mentioned, has no defined right answers. Its efficiency is rarely considered.

2.5 Measuring ideation

For lack of objective measures (beyond patentability), currently ideation with AI is using linguistic-like tools. It is assessed by quantity, quality, (linguistic) novelty and (linguistic) variety (Shah et al., 2003), or by novelty and usefulness together (Sarkar and Chakrabarti, 2011). But a brainstorm is worth its best ideas, since the rest are discarded: idea-generation processes should be compared on the quality of the best idea, not the average (Girotra et al., 2010), and extreme-value models describe how the best of many concepts improves with the number tested (Dahan and Mendelson, 2001). Mean scores, the usual summary in LLM ideation studies, hide exactly this difference. To avoid this trap, here we combine the tail and the cost in one measure, top ideas per token, and vary only how the product is presented, holding the model, the method and the prompt fixed.

3 Methods

Two experiments are conducted. Experiment 1 (RQ1–RQ3) holds the model and the prompt fixed and varies two inputs across all 13 market needs that the graph can serve: how much of the product graph the prompt contains (the *representation*) and how long the model may think (the *effort*). It measures what each combination produces and what it costs. Experiment 2 (RQ4) fixes the prompt, the effort and the market need (M6), asks for 50 ideas per call instead of 5, and varies the representation and the generator model (Haiku, Opus and Fable). It tests

whether the effect of the representation holds across models, and what each model costs per top idea. Table 1 gives an overview of Experiment 1 and the subsections below give its details; Experiment 2 is described in Section 3.9.

Table 1: Experiment 1 (RQ1–RQ3) at a glance. Experiment 2 is summarised in Table 6.

Item	Value	Details
Product graph	Hand-built innovation matrix of a microwave oven: 123 nodes, 246 typed edges	Section 3.1, Table 2
Market needs	13 of 17; four (M2, M9, M10, M15) are addressed by no function, so their slice is empty	Section 3.2
Representations	**A** full graph; **B** vertical slice of the target market need; **R** random slice with B's number of nodes of each type; **C** no graph (target market need only)	Section 3.2, Table 3
Task	SUBSTITUTE operator, one worked example, 5 ideas as JSON; the prompt differs across conditions only in the graph file	Section 3.3, Appendix B
Generator	`claude-haiku-5-5`, default sampling; effort low, medium, high or xhigh	Section 3.4
Design	5 replicate calls per cell; A and B at low–high on 13 needs and at xhigh on 6; R and C at high on 13; 580 calls, 2,892 ideas	Table 4
Judge	`claude-sonnet-5-5`, medium effort, blind to condition and effort	Section 3.5
Set judging	Pairs of idea sets differing in one factor, judged in both orders; 1–5 on five criteria	Table 5
Idea scoring	Every idea rated 1–5 on four criteria; score = mean; a *top idea* scores ≥ 4.0 (12.3% of ideas)	Sections 3.5, 3.6
Statistics	Pairwise: win rate (ties = half), Wilson 95% CI, exact sign test, 45% non-inferiority margin for B vs A. Top tail: market-need-averaged rates, bootstrap 95% CI (2,000 resamples of calls within need)	Section 3.7

3.1 Product graph

The innovation matrix. The innovation matrix (IM) of Leo Tchirsky, as taught in design-thinking workshops (Berengueres, 2023), describes a product as stacked layers: the **market needs** it serves, the **functions** that serve them, the **components** that perform the functions, the **technologies** that make the components possible, and the scientific **fields** behind those technologies (Figure 1). Reading upwards answers *why* a part exists; reading downwards answers *how* a market need is met. Innovation happens by changing a link: a new technology enables a different component, a component takes over another's function, or a function is found to serve a market need nobody had linked it to. In a workshop the matrix is drawn on a wall; here it is a typed graph that a model can read.

The microwave matrix. The product is a countertop microwave oven, the simplest mechanical model: a timer knob, a power knob, a door button and an interior lamp. The authors built its IM by hand in a single YAML file (`microwave_base_graph.yaml`), adding two cross-cutting types the workshop matrix keeps beside it, constraints and actors. It has 123 nodes and 246 directed, typed edges (Figure 2; all nodes listed in Appendix A).

Nodes belong to four stacked layers and four cross-cutting types (Table 2):

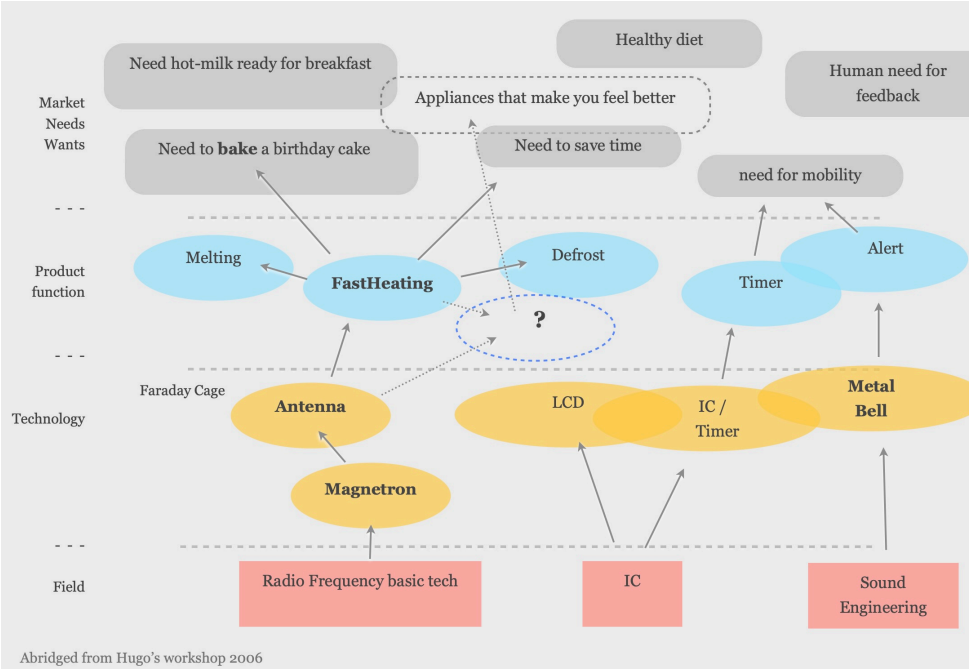


Figure 1: An innovation matrix as drawn in the workshop material (Berengueres, 2023).

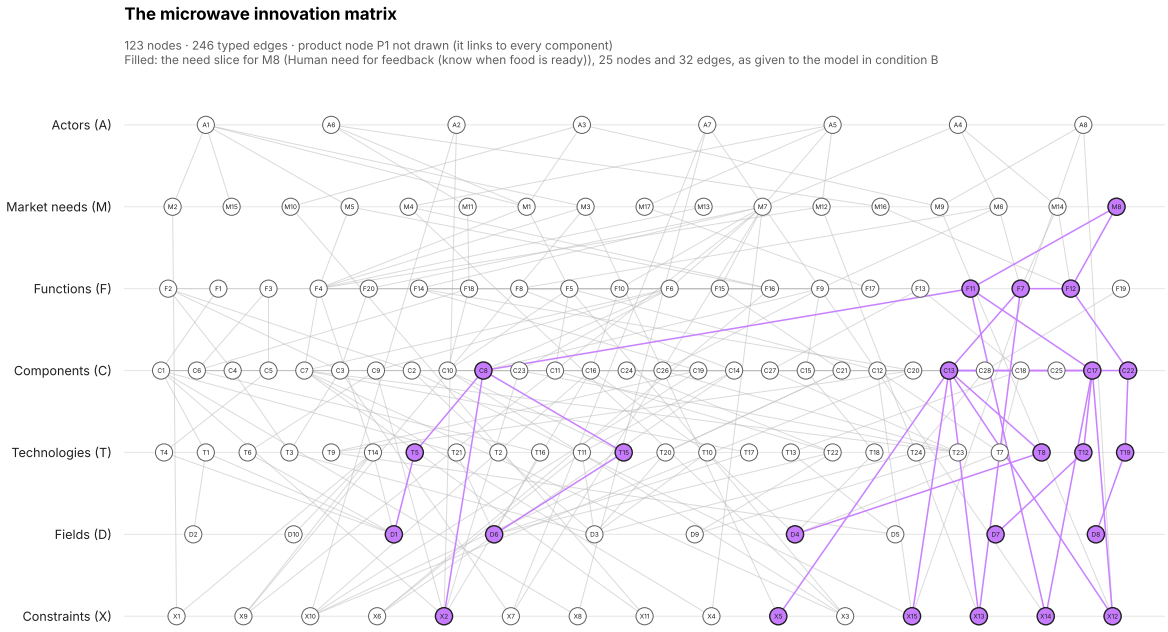


Figure 2: The microwave innovation matrix used in the experiment. Rows are the layers, bottom to top, with constraints below and actors above. Filled nodes and edges show, as an illustration, the market-need slice for M8, *know when food is ready* (25 nodes, 32 edges): what the model sees in condition B for that need, against the whole graph in condition A. M8 was chosen for illustration. It overlaps with the prompt's worked example (seeing the food) and is one of the needs where B's advantage over A is largest (Figure 3 shows all 13 slices).

Table 2: Node types in the microwave innovation matrix (123 nodes): count per type and what each type represents.

Type	Count	Role
MarketNeed (M)	17	what users want (e.g. M4 *Defrost frozen food quickly*)
Function (F)	20	what the product does
Component (C)	28	physical parts
Technology (T)	24	the principles that enable components and functions
Field (D)	10	scientific fields the technologies are rooted in
Constraint (X)	15	limitations and conflicts (e.g. X3 *uneven heating*)
Actor (A)	8	user groups (e.g. A5 *parent with young children*)
Product (P)	1	the microwave oven

Edges carry one of twelve types. Those that build the layer stack are `has_component`, `performs_function`, `enables_function`, `addresses_need`, `enabled_by` and `rooted_in`. Component-level flows are `feeds` and `actuates`. Constraint links are `causes`, `contradicts` and `partially_mitigates`. Weak associations are `relevant_to`. A thirteenth type, `resolves`, is declared in the schema for agents to use when a new node replaces or solves an existing one; it does not occur in the base graph.

Each node has an identifier, type, label and a one-line detail. The file begins with a meta block that names the layers and defines the edge types.

3.2 Representations

Figure 3 shows what the model sees in each condition. Every prompt contains a graph file produced by one serializer from the base graph. Fields, ordering and the meta block are the same in every file; only *which* nodes and edges are present differs. No file states which condition it belongs to. For each target market need M_k (`slicer.py`):

- **A, full graph.** All 123 nodes and 246 edges. One file shared by all market needs.
- **B, market-need slice.** The vertical slice of M_k, obtained by walking down the matrix from the market need:

  1. the functions that `addresses_need` M_k, plus every function that enables them through `enables_function`, recursively;
  2. the components that `performs_function` one of those functions;
  3. the technologies that enable those components or functions (`enabled_by`);
  4. the fields those technologies are `rooted_in`;
  5. every constraint linked to any node above by `causes`, `contradicts`, `partially_mitigates` or `relevant_to`, and every actor `relevant_to` M_k;
  6. the product node.

  The file contains the edges induced among the selected nodes.
- **R, random slice.** The target market need and the product, plus a random draw of nodes with the same number of nodes of each type as B, with the induced edges. The draw is seeded by the market need identifier (`random.Random("<need>-1")`), so it is reproducible and independent of run order. R controls for the *size* and layer composition of B while removing its *relevance* to the market need.
- **C, no graph.** The meta block (layer names and edge-type definitions) and the target market need node only.

Four needs (M2, M9, M10, M15) are not addressed by any function, so their vertical slice is empty. They were excluded, leaving 13 target market needs: M1, M3, M4, M5, M6, M7, M8, M11, M12, M13, M14, M16, M17. Table 3 gives the size of each representation.

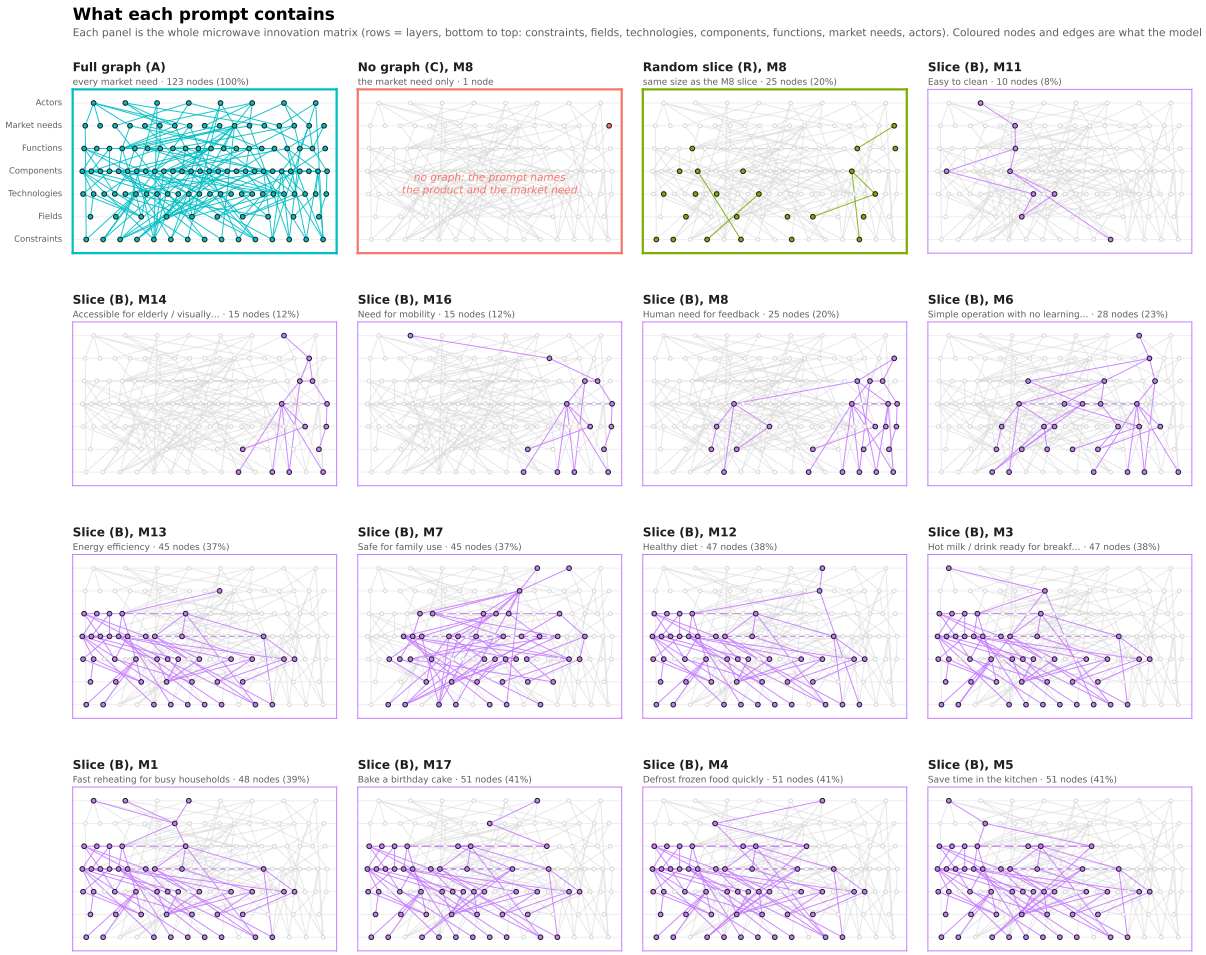


Figure 3: What each prompt contains. Every panel is the whole microwave innovation matrix (rows are the layers, bottom to top: constraints, fields, technologies, components, functions, market needs, actors); coloured nodes and edges are what the model sees. Top row: the full graph (A), no graph (C, only the market need), a random slice of the same size as the M8 slice (R; M8 is the illustrative need of Figure 2; it is not representative, see its caption), and the smallest market-need slice (M11). The other panels are the market-need slices (B) for the remaining 12 needs, ordered by size (10 to 51 nodes, 8–41% of the graph).

Table 3: The four representations of the graph given to the model: size of the graph content in the prompt and mean input tokens per call. Ranges for B and R are across the 13 market needs.

Condition	Nodes	Share of graph	Edges	Input tokens per call (mean, incl. prompt)
A	123	100%	246	10,912
B	10–51 (median 45)	8–41%	11–85	3,778
R	10–51, same as B for each market need	8–41%	2–52	3,143
C	1	1%	0	1,055

R matches B in node count and type composition but not in edges. Randomly drawn nodes are less connected to one another than the nodes of a vertical slice, so R contains fewer induced edges than B for every market need. It also uses 17% fewer input tokens. R shares 3–29 nodes with B for the same market need, mostly by chance among the larger slices. A manifest (`slices/manifest.csv`) records, for every file, the node identifiers, counts per type, edge count, token count and SHA-256 hash.

3.3 Task and prompt

All calls use one prompt template (`template_prompt.txt`, Appendix B). It asks the model to innovate the product "based on the graph given", encourages radical and wild ideas, and asks it to favour innovations that serve the target market need. Placeholders are filled by literal substitution: product name, target market need identifier and label, operator name and definition, the graph file, and the number of ideas.

- **Operator.** The *Substitute* operator from SCAMPER (Eberle, 1971) in every call: "Identify components and the functions they perform, then replace them with something cheaper or simpler that fulfils the same function equally well or better."
- **Worked example.** One example of an innovation the brief likes, taken from the authors' workshop material (Berengueres, 2023). The door's mesh and the interior lamp both serve the user's need to see the food, so they can be replaced together by a webcam, or by a transparent casing that needs no lamp.
- **Output.** Exactly 5 ideas as JSON. Each idea has a title; the identifiers of existing nodes it was applied to (*anchors*); new nodes (type, label, one-line detail); new edges; and a one-sentence rationale for how it serves the target market need.

The prompt is identical across conditions except for the graph file. In condition C, the "graph given" is only the schema and the target market need.

3.4 Generation

- **Model.** `claude-haiku-5-5`, through the Anthropic Messages API, with streaming. No sampling parameters were set; the model does not accept a temperature setting, so all calls use its defaults.
- **Thinking effort.** Set per call through the API's effort parameter, at four of the five available levels: low, medium (the model's default), high and xhigh. Thinking tokens are billed and reported as output tokens.
- **Output limit.** 48,000 tokens (64,000 for xhigh). No call reached its limit.
- **Design.** Five replicates per cell, each a fresh call with the identical prompt:

Table 4 summarises the design.

Table 4: Generation design: conditions run at each thinking effort and on how many market needs, with 5 replicate calls per cell (580 calls in total).

Conditions	Efforts	Needs	Calls
A, B	low, medium, high	13	390
A, B	xhigh	6 (M1, M4, M6, M8, M13, M17)	60
R, C	high	13	130
Total			**580**

The xhigh cells cover 6 market needs, chosen to span small and large slices, to limit cost. R and C were run at high effort only.

- **Parsing.** A response is valid if it parses as JSON with an `ideas` list. 579 of 580 calls were valid: 576 returned 5 ideas and three returned 6, 3 and 3 ideas; one returned no parseable ideas and is kept as a failure. In total, the calls produced 2,892 ideas.
- **Logging.** The runner appends one JSON line per call to an append-only log (`runs/rq1_full.jsonl`). Each line holds the run identifier (condition, market need, effort, replicate), the graph file's identifier and hash, the prompt hash, model, output limit, input and output tokens, latency, stop reason, the parsed ideas and the raw response. Runs are resumable, and cells can be added later without rerunning earlier ones.

3.5 Judging

Ideas were judged blind by a second, larger model, `claude-sonnet-5-5`, at medium effort, in two ways. In both, the judge saw each idea's **title, new elements (label and detail) and rationale only**. Node identifiers, anchors, edges, condition, effort and run identifiers were removed, so nothing revealed which graph the idea came from. The brief shown to the judge gave the product, target market need, operator, and the instruction that radical, wild ideas serving the market need were wanted.

Pairwise set judging. Two idea sets for the same market need and replicate, differing in one factor, were shown side by side. The judge scored each set from 1 to 5 on relevance, novelty, feasibility, specificity and diversity (how different the ideas are from each other), and named the better set or a tie (prompt in Appendix C). Every pair was judged twice, once in each presentation order, and the two verdicts were combined into one result per pair: agreeing verdicts count as a win, opposite verdicts as a tie. If one side had no parsed ideas, the other won without a judge call (one pair per affected comparison). The comparisons judged were:

Table 5 lists the comparisons judged.

Table 5: Pairwise set comparisons sent to the judge. Each pair is two idea sets for the same market need and replicate that differ in one factor.

Comparison	Effort	Pairs
A vs B	low, high	65 each
B vs R, B vs C, A vs C	high	65 each
high vs low, within A and within B	—	65 each
xhigh vs high, within A and within B	—	30 each
B at high vs A at low	—	65

Individual idea scoring. Each call's five ideas were sent to the judge in one request, with the instruction to rate each idea on its own merits, independently of the others. Each idea was scored from 1 to 5 on relevance, novelty, feasibility and specificity (prompts in Appendix D). An idea's **score** is the mean of the four ratings. All 2,892 ideas were scored; two calls whose first

scoring returned the wrong number of ratings were scored again.

3.6 Measures

Cost. Input, output and total tokens per call, from the API's usage report.

Use of the graph (computed without model calls, from the parsed ideas):

- *anchors outside the market-need slice*: the share of anchors that are existing nodes outside B's slice for that market need;
- *anchors not in the graph*: the share of anchors that match no node identifier in the base graph;
- *ideas echoing the worked example*: ideas whose title or new nodes mention a webcam, camera or transparency;
- *new nodes per idea*, and *words per idea* in the text the judge sees.

Top tail. A **top idea** is an idea whose individual score is at least 4.0, that is, one the judge rates 4 or higher on average across relevance, novelty, feasibility and specificity. The threshold is a fixed point on the judge's scale, not a percentile, so the share of ideas that clear it depends on what is being judged. Scores move in steps of 0.25 (one step of a 1–5 rating, averaged over four criteria), so no exact percentile cut exists; in Experiment 1, 4.0 is the step closest to the top 10% and keeps 12.3% of all ideas (355 of 2,892). As a stricter check, 4.25 keeps the top 2.4% (68 ideas).

What "top" means, and what it does not. A top idea is top *relative to this setup*: one judge model, one rubric, one brief, one product and one operator. It is not a global benchmark of idea quality or of innovativeness. A score of 4.0 says the judge found the idea relevant, plausible and concrete for this brief; it does not say the idea is new to the world (Section 3.10 shows that most top ideas restate existing technology), and it cannot be compared with scores from other judges, rubrics or studies. Within the study, the fixed threshold makes arms comparable: an arm that produces more ideas the judge rates highly gets more top ideas. A percentile cut per arm or per model would instead give every arm the same share by construction and hide exactly the differences being measured.

Target market needs differ widely in how easily they yield top ideas (from 3% to 29% of ideas at 4.0), and the xhigh arms cover only 6 market needs, so every rate is computed per market need and then averaged over the market needs an arm was run on, each market need weighing the same. For each arm:

- **top-idea rate**: the market-need-averaged share of the arm's ideas that are top ideas;
- **calls with a top idea**: the market-need-averaged share of calls producing at least one top idea;
- **tokens per top idea**: the arm's total tokens divided by its number of top ideas;
- **chance of a top idea for a budget** b: the probability of at least one top idea when b tokens are spent on whole calls of that arm, averaged over market needs. For need j, with p_j the share of calls containing a top idea and $\bar{t}_j$ the mean tokens per call, $k = \lfloor b/\bar{t}_j \rfloor$ calls succeed with probability $1 - (1 - p_j)^k$. Calls are treated as independent. Because 5 calls per market need often give a raw share of exactly 0 or 1, p_j is the Jeffreys estimate $(h_j + \frac{1}{2})/(n_j + 1)$, where h_j of n_j calls contained a top idea.

Mean idea score. The market-need-averaged mean individual score per arm, and the mean of each criterion.

3.7 Statistics

Pairwise judging. The unit is the pair, not the verdict: the two presentation orders of a pair are not independent, so they are combined first. For each comparison:

- the *win rate* of the first-named arm, with ties counting as half a win;
- a Wilson 95% confidence interval;
- an exact two-sided sign test on wins against losses, ties excluded;
- the *order agreement*, the share of pairs that received the same verdict in both orders;
- the mean score difference per criterion.

Whether the market-need slice is no worse than the full graph is tested as non-inferiority with a margin of 45%: B is non-inferior if the lower bound of its win-rate interval against A is above 45%.

To check whether judges favour longer sets, each pairwise record stores the number of words each side showed the judge. The share of decided pairs won by the longer set is reported next to each win rate.

Top tail. 95% confidence intervals for the top-idea rate and the call rate come from 2,000 bootstrap resamples. Each resample redraws calls with replacement within each market need, which keeps the five ideas of a call together and the market-need-averaged weighting fixed. The intervals treat each idea's score as fixed. They capture variation between generation calls but not the within-item variability of the judge, which can assign a different score to the same idea on repeated evaluation; ideas scored near the threshold could then fall on either side of it. The intervals are therefore likely too narrow. The 4.0 threshold was set after the scores were known (it is the step closest to the top 10% of Experiment 1's ideas), so it is a post hoc choice.

No correction for multiple comparisons was applied. Per-market-need results rest on 5 calls or pairs each and are reported only as patterns.

3.8 Pilot and plan (Experiment 1)

A pilot (2 market needs, M8 and M16; A and B; low, medium and high effort; 3 replicates; 36 calls) tested the pipeline before the full run. Half of its high-effort calls (6 of 12) used up the 8,000-token output limit thinking and returned no ideas. Its 17-of-18 advantage for the full graph did not hold at full scale (Section 4.7.2).

- **Planned before the full run.**
  - The four conditions, including R and C as controls.
  - The A vs B comparison at low and high effort, with the hypotheses that the full graph yields better sets and that B is non-inferior to A at a 45% margin.
  - The effort comparison, high vs low.
  - The hypotheses that B beats R and that a graph beats no graph, to be tested by pairwise judging.
- **Added after the pilot.**
  - An output limit of 48,000 instead of 8,000 tokens to accommodate longer responses for high and xhigh effort; streamed calls, which the API requires for calls that long; and effort in the run identifier, so that cells could be resumed.
  - Medium and xhigh effort in the full run.
  - The B-high vs A-low comparison.
  - Judging every pair in both orders, the length check, and individual idea scoring, which was first used to rank ideas for inspection.
- **Exploratory, defined after seeing the results.**
  - The top-tail analysis: the 4.0 threshold, top-idea rate, tokens per top idea and chance for a budget.
  - The market-need-averaged mean idea scores.

The top-tail measures follow from the purpose of brainstorming (Section 1), but they were chosen after the pairwise results were known and are reported as exploratory. Their conclusions about relevance (B against R) are not confirmed by the planned pairwise test (Section 4).

3.9 Experiment 2: model comparison (RQ4)

Experiment 1 asked for five ideas per call, and read one by one those ideas converge on the same few moves whatever the representation and the effort: asked for five, the model returns its five most obvious answers. Experiment 2 asks for 50 ideas per call on a single market need, which forces the model past the obvious ones, and repeats the run with three generator models. It asks whether the effect of the representation found in Experiment 1 holds across models, and what each model costs per top idea. Everything not described here is as in Experiment 1: the product graph, the four representations and their files, the operator, the worked example, the judge and its prompt. Table 6 gives an overview.

Table 6: Experiment 2 at a glance.

Item	Value
Market need	One: M6, *simple operation with no learning curve* (slice: 28 nodes, 23% of the graph)
Representations	A, B, R and C, the same files as in Experiment 1 (R: the same random draw)
Task	As in Experiment 1, except "Return exactly 50 ideas" instead of 5
Generators	`claude-haiku-5-5`, `claude-opus-5-5`, `claude-fable-5-1`; medium effort; output limit 48,000 tokens
Design	10 calls per representation and model: 40 calls per model, 120 in total, in random order
Judge	`claude-sonnet-5-5`, medium effort, the prompt and criteria of Experiment 1, blind to representation, model and position in the list
Measures	Yield per 100k tokens (as in Experiment 1) and per US dollar at list price; top-idea rate
Statistics	Bootstrap 95% CI, 2,000 resamples of calls within each arm

Choice of market need. The need was chosen before any 50-idea call was run, by applying four rules to the 13 needs of Experiment 1, in order:

1. *No overlap with the worked example.* The webcam and the transparent casing must not themselves answer the need. This excludes M8 and M16.
2. *Every representation found something.* Each representation must have produced at least one top idea for the need at high effort in Experiment 1. This excludes M1, M5 and M11.
3. *Same order as the pooled result.* The representations must rank as in the pooled Experiment 1 result at high effort (B > C > R > A). This excludes M3, M4, M7, M12 and M17.
4. *Chosen by the authors from the rest* (M6, M13, M14). M6 was chosen because operating the appliance is a central, long-standing usability problem that no single technology dominates.

Prompt. The prompt is Experiment 1's word for word, except for the one line that sets the number of ideas. This was checked by restoring "5" in each Experiment 2 prompt and matching the prompt hash logged in Experiment 1. The persona sentence is identical; only its name in the run identifiers differs, so that the two experiments' runs cannot be confused.

Generation. Each model received the same 40 prompts in the same random order, at medium effort, through the Anthropic Messages API with streaming and default sampling. Effort was fixed at medium for four reasons. It is the default for all three models, so it is the setting a practitioner would use without tuning. Experiment 1 showed that effort raises quality but not yield: the market-need slice was most efficient at medium (8.8 top ideas per 100k tokens) and fell at high and xhigh effort (Section 4.5). A 50-idea call already writes 15,000–25,000 output tokens,

and higher effort would multiply the thinking billed as output (about twice the output tokens from medium to high, and about five times to xhigh, in Experiment 1), which at Opus and Fable prices would have made the run several times more expensive for no expected gain in efficiency. Finally, holding effort fixed leaves representation and model as the only factors that vary. The same effort label need not buy the same amount of thinking in every model; Experiment 2 therefore compares the models as they behave at their default, not at equal thinking. All 120 calls returned valid JSON and ended normally; none reached the output limit. Asked for 50 ideas, the models returned between 44 and 60 per call (Haiku 44–60, Opus 49–52, Fable 50–52); every idea returned is used. Identical prompts produced identical input token counts across the three models (for example 10,910 tokens for every full-graph call), so a token is the same unit for all three and per-token yields can be compared across models.

Table 7: Experiment 2 generation by model: calls, ideas returned and scored, tokens and cost at list price.

Model	Calls	Ideas	Input tokens	Output tokens	Cost (USD)
Haiku 5.5	40	2,080	173,120	671,711	0.35
Opus 5.5	40	1,990	173,120	631,352	13.32
Fable 5.1	40	2,021	173,120	583,199	30.89

Judging. Every idea was scored individually by the judge of Experiment 1, with the same prompt and the same four criteria. A 50-idea list was sent to the judge in requests of five consecutive ideas, so that every request had the form of an Experiment 1 request. The judge saw each idea's title, new elements and rationale, and nothing that revealed the representation, the model or the idea's position in its list. Of 6,091 ideas, 6,066 were scored; 25 (0.4%: 5 from Opus, 20 from Fable) could not be scored after a retry and are excluded, while their calls' tokens are counted.

Measures.

- **Top idea:** an idea scoring at least 4.0, the same fixed threshold as in Experiment 1. Because the threshold is a point on the scale and not a percentile, the share of ideas clearing it differs by model: 6.1% of Haiku's ideas (126 of 2,080), 13.6% of Opus's (270 of 1,985) and 14.7% of Fable's (295 of 2,001), against 12.3% in Experiment 1. These shares are the quantity compared: they show how often each model's ideas reach the level the judge rates highly. They are not comparable with Experiment 1's, because 50-idea lists have shorter ideas, which the judge scores lower, and they inherit the judge's limits: Sonnet may reward polished wording. A content-only re-rating of the extremes (Section 4.6.4) bears this out for Fable against Opus but not for the larger models against Haiku.
- **Top-idea rate:** the share of an arm's ideas that are top ideas.
- **Yield per 100k tokens:** an arm's top ideas divided by its input and output tokens, as in Experiment 1. All ideas and tokens of a call are counted.
- **Cost and yield per dollar:** each call's cost is its input and output tokens at the provider's list price per million tokens, as published on 20 October 2026: Haiku 5.5 \$0.10 input and \$0.50 output, Opus 5.5 \$4 and \$20, Fable 5.1 \$10 and \$50. Thinking tokens are billed as output. No batch or caching discount is applied. The yield per dollar is an arm's top ideas divided by its cost. The judge's cost is the same for every model and is not included.

Statistics. With one market need, the bootstrap resamples the 10 calls of each arm with replacement (2,000 resamples), keeping each call's ideas and tokens together. Arms are compared by these intervals. The distribution of single-call yields is shown alongside as a description of how much individual calls vary, not as a test. No correction for multiple comparisons is applied.

Plan and exploration. The Haiku run was planned and its hypotheses recorded on 2026-10-19, before any 50-idea call. Those hypotheses concern how many distinct ideas each repre-

sentation reaches and whether later ideas stay on the need; they require coding every idea into a codebook of moves, which is in progress and not used here. The Opus and Fable runs were added on 2026-10-20, after the Haiku scores had been seen, and the yield per dollar after that. All RQ4 results are therefore exploratory.

Limitations specific to Experiment 2.

- *Scores are not comparable with Experiment 1.* Asked for 50 ideas, the models write shorter ideas (about 35 words per idea for all three models, against about 100 in Experiment 1), which the judge scores lower, so top-idea rates and yields are compared within Experiment 2 only.
- *The judge is weaker than two of the generators.* Sonnet judges ideas from Opus and Fable, and may not recognise the best ideas of a stronger model. Judge and generators also come from one model family.
- *Scope.* One product, one market need, one effort level and 10 calls per arm.
- *Prices change.* List prices are a snapshot. The ranking per dollar would change with discounts or new prices, while the yield per token would not.

3.10 Sample of generated ideas

One of the authors read all the ideas from Experiment 1, which led to Experiment 2. LLM-ideation studies usually assess novelty with embedding measures such as the semantic distance between ideas, which show how the ideas differ from each other but not from prior art. Read against what is already known in the domain, three things stood out:

1. **The top tier is state of the art, not invention.** Two ideas from the judge's top decile from Experiment 1:

   `A_M13_high_r03` — *Solid-State RF Core: Magnetron and HV Stack Replaced by GaN Amplifiers*
   `B_M13_low_r01` — *Magnetron-free solid-state RF cooking head*

   As engineering these are sound; as innovation their value is close to zero. Gallium-nitride power semiconductors made it practical, in the early 2020s, to retire the noisy magnetron and convert the whole RF subsystem to solid state, with gains in weight, noise and vibration, and in the precision of power delivery and control. Dozens of patents apply GaN to microwave ovens and the first commercial products exist; the open problem is cost, at present roughly ten times the magnetron subsystem. A prior-art check would score these ideas at the floor, where the judge's novelty criterion put them near the ceiling.

   The middle of the ranking tells the same story. Five ideas drawn from around the median:

   Sliding RF-Choke Curtain Replaces the Hinged Door and Push Button;
   Photoluminescent cavity glow replaces the 20–25 W lamp;
   Sputter-Listening Auto-Stop replaces the timer;
   Acoustic Thaw: Ultrasonic Pre-Defrost Replacing Low-Power Cycling;
   Phased-Array Field Sweep (No Turntable).

   Every one of them is prior art or commercially available: sensor-based auto-stop has been standard in consumer ovens for decades, solid-state ovens already steer the field electronically instead of rotating the food, and ultrasound-assisted thawing is established in food processing. From the top of the ranking to its middle, the judge orders the state of the art; it does not separate the new from the known.

2. **Scores are biased towards fit.** A sample from the bottom decile:

Ghost Lamp Vision: Replace the Interior Lamp With Microwave-Transparent Walls;
GaN Universal-Mains Power Stage Replaces HV Transformer Stack;
Capacitor-Free 48 V Bus Power Stage;
Direct-Coupled Magnetron Pod (No Waveguide Duct).

The GaN power stage here is not materially different from the GaN ideas at the top of the ranking. Part of the gap is defensible, since the judge marks a power-supply substitution down on relevance when it is aimed at a market need it cannot serve; but the same technical move appearing at both ends of the ranking shows that an AI ranker is not entirely consistent, and that its scores reflect the fit between a move and the target market need as much as the move itself.

3. **Diversity.** A topic analysis (Nvivo; not shown) finds the same ideas concentrated in a few recurring topical pathways: solid-state GaN RF generation, transparent or lamp-free enclosures, simplifications of the RF chain, and sensor-based doneness feedback. This should not surprise anyone: each call asked for five ideas under one operator, and further calls revisit the same pathways more often than they open new ones.

Karpathy has described a language model as "a lossy compression of the internet" (Karpathy, 2023), and the compressed corpus includes the patent and product literature of this domain. Read this way, the generator behaved as a retrieval system: excellent at surfacing the engineering moves that bear on a market need, which is also why the ∼11,000-token full-graph prompt and the ∼1,000-token no-graph prompt converge on the same few pathways. This matches the argument that language models excel at induction and increasingly at deduction, but not at abduction, the leap to new premises that genuine invention requires (Zahavy, 2026). The setup cannot be blamed for presenting the state of the art as innovation, if innovation is defined as what no one has patented before. What the experiment measures is therefore the efficiency with which a representation surfaces strong, relevant engineering moves, the raw material of ideation, not patentable novelty; the same reading applies to published systems whose "innovative" output is rated rather than checked against prior art (Section 2.3).

The compression is of what has been written, and what is written about microwave ovens is mostly technology. Reading all the ideas, what is absent is as telling as what is present. Cost appears nowhere as the object of innovation, although the graph carries an explicit low-cost constraint (X12) and cost is the main force selecting microwave innovations: the magnetron has outlived every technically superior alternative because fifty years of manufacturing scale have made it extremely cheap, and that advantage is self-reinforcing, a path dependency that any real substitution must pay its way out of. Nothing in the ideas reflects the open-source hardware movement and its community-designed and community-repaired appliances, and nothing uses the fact that most households run a microwave next to a conventional oven, so that several market needs, crisping and browning among them, are already served by the appliance beside it. This knowledge lives in markets, communities and kitchens more than in patents and specification sheets, so the lossy copy retrieves the recorded state of the art and misses the selection environment that decides which of those ideas becomes an innovation.

3.11 Reproducibility

All code, product graphs, raw call traces, and judge outputs are available at `https://github.com/orioli/IM-LLM`. The artifact supports two reproducibility modes:

1. **Offline re-analysis (zero-cost):** The complete raw dataset of Experiment 1's 580 generation calls (2,892 ideas) with prompt hashes, raw outputs, and token counts (`runs/rq1_full.jsonl`), together with pairwise and individual judge ratings (`results/`),

is bundled in the repository, as are Experiment 2's 120 calls and their scores, one run log and results folder per model (`M6_50ideas/`). All statistical tests, bootstrap intervals, tables, and figures can be regenerated locally offline without API keys or costs.

2. **Live pipeline execution:** Slicing, prompt construction, API execution harnesses, and evaluation scripts are parameterized to rerun the study from scratch or adapt it to other models.

Generation consumed 8.6 million tokens in Experiment 1 (3.6 million input, 5.0 million output, all `claude-haiku-5-5`) and 2.4 million in Experiment 2 (0.5 million input, 1.9 million output, across the three models; $44.56 at list prices). `claude-sonnet-5-5` was used for all judging. Appendix E lists the repository structure and artifacts.

4 Results

4.1 Summary

The aim of brainstorming is the rare great idea, not a good average, so the main measure here is the **top tail**: how often an arm produces a top idea (score ≥ 4.0, the top $\sim$12% of Experiment 1's ideas), and what each such idea costs.

RQ1. Which representation yields the most top ideas per token? (Section 4.3)

- **A relevant slice of the graph finds top ideas fastest per token.** Counted against tokens spent, top ideas accumulate in a straight line for every arm, so each arm has a yield: top ideas per 100k tokens. The market-need slice (B) at medium effort yields 8.8, about three times the full graph (A) at any effort (2.6–3.2). B's top-idea rate is also the highest at equal effort (15% at medium and high, against 10% for no graph and for a random slice of the same size). In Experiment 2 (RQ4), where every representation ran at the same effort, the slice matched but did not beat no graph, for all three models.
- **The efficiency gain holds market need by market need, and is largest for small slices.** The slice is more efficient than the full graph on 10 of 13 market needs (sign test $p = 0.039$), and no graph beats a random slice on 10 of 12. The slice's lead is 3–5$\times$ for market needs whose slice has 15–28 nodes, and 1–2$\times$ for those with 45–51 nodes (exploratory).
- **No graph is surprisingly efficient.** Its calls are the cheapest, so despite a low top-idea rate its yield (6.4) beats the random slice (4.1) and the full graph (3.1), and is close to the slice at high effort (5.9).

RQ2. Does the amount or the relevance of the graph matter? (Section 4.4)

- **The full graph (A) writes the best idea *sets* but does not find more top ideas.** Judged as whole sets, A beats B 64% of the time and no graph 74% of the time. But its top-idea rate is no higher than B's, and its yield is about 3 top ideas per 100k tokens at every effort, half or less of B's: its advantage is the specificity of an average set, not more standout ideas.
- **Relevance vs size is only partly settled.** In the top tail, the market-need slice beats the size-matched random slice (top-idea rate 14.7% vs 9.8%; yield 5.9 vs 4.1 per 100k tokens; the slice's slope is higher in 99% of paired bootstrap draws). Judged as whole sets, the two cannot be told apart (B over R 54%, $p = 0.55$). What a relevant slice adds shows up in the best ideas, not in the average set.
- **A graph helps, clearly so when it is the full graph.** A beats C on whole sets (74%, $p < 0.001$); B beats C less clearly (62%, $p = 0.053$) but has a higher top-idea rate (14.7% vs 10.2%).

RQ3. How does spending tokens on context compare with spending them on thinking effort? (Section 4.5)

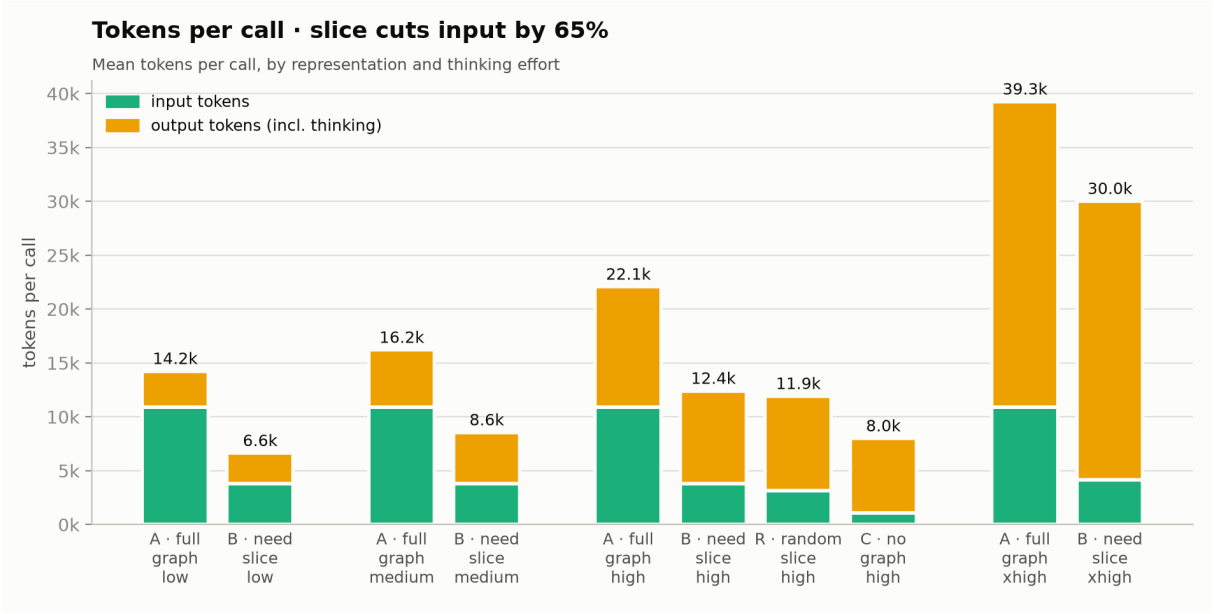


Figure 4: Mean tokens per call by representation and effort, split into input (the prompt, most of it the graph file) and output (thinking and answer). The full graph pays for its context on every call; thinking effort drives the output for every representation.

- **Thinking effort is the strongest lever on quality, but not on yield.** High beats low in about 93% of set comparisons, and xhigh produces the highest top-idea rates of all (19–21%). But each call costs proportionally more, so yield does not rise with effort: for the full graph it stays at about 3 per 100k tokens from low to xhigh, and for the slice it peaks at medium (8.8) and falls at high (5.9) and xhigh (3.1).

RQ4. Does the effect of the representation hold across models, and what does each model cost per top idea? (Section 4.6)

Asked for 50 ideas per call on market need M6, all three models had their lowest yield with the full graph (clearly so for Haiku and Fable; for Opus the interval only just separates from the slice's and overlaps with no graph and the random slice): per token, its yield was about half that of the slice or no graph for Haiku (9.4 against 20.4 and 22.2 top ideas per 100k tokens) and Fable (26.4 against 51.3 and 53.6), and the lowest for Opus (26.3 against 39.9 and 34.8). The slice and no graph were indistinguishable for every model. For Opus and Fable the full graph did not lower the share of top ideas; it only cost more. The stronger models found roughly 1.5 to 2.5 times as many top ideas per token as Haiku, but at list prices Haiku found about 20 times more per dollar than Opus and about 40 times more than Fable. Exploratory.

4.2 Cost and validity (Experiment 1)

The design is summarised in Table 1 and detailed in Section 3; this section reports what Experiment 1's calls cost and how many responses were valid. Experiment 2's cost and validity are reported in Section 3.9 (Table 7), and its plan in the same section.

Figure 4 shows tokens per call by condition and effort; Figure 5 sets the four representations side by side at high effort, input tokens against mean idea score.

Table 8 gives the means per call.

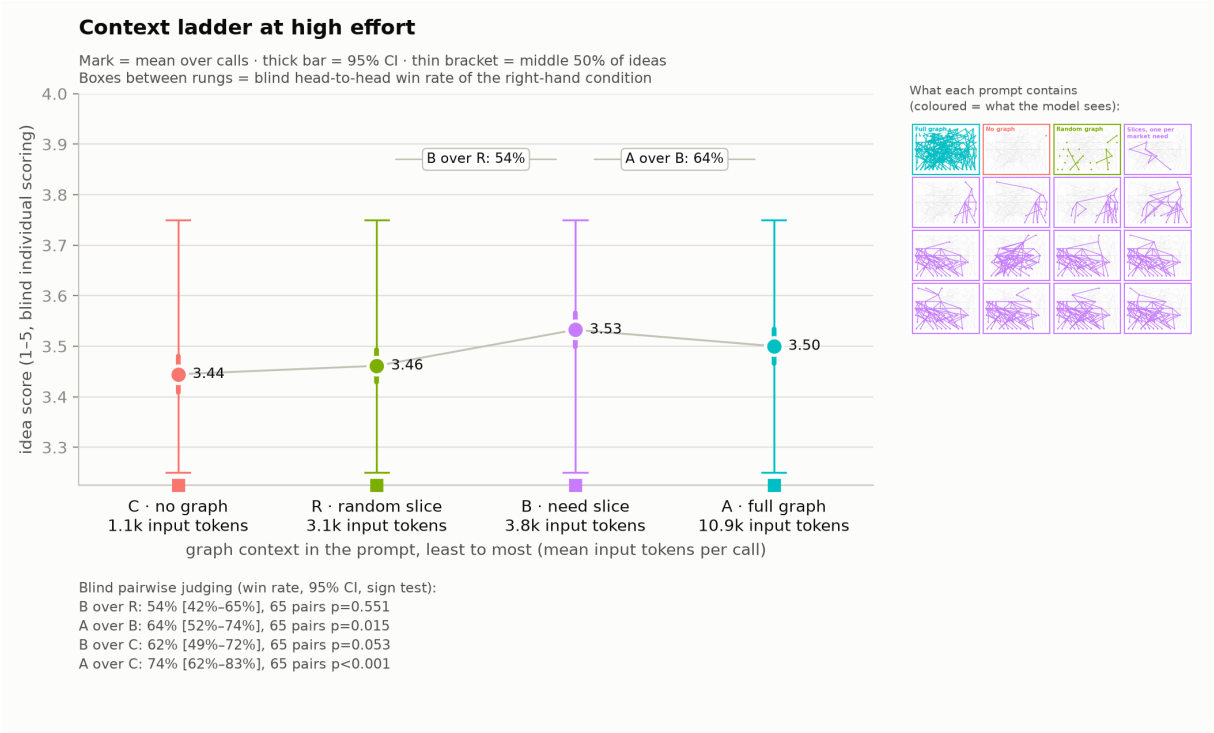


Figure 5: The context ladder at high effort: no graph (C), random slice (R), market-need slice (B) and full graph (A), in order of input size, against mean idea score, with the head-to-head set win rates between rungs. Mean score is shown for reference; the main measure is yield, top ideas per token.

Table 8: Cost and validity by representation and effort: number of calls, share of responses parsed, mean tokens per call (input, output, total) and mean words per idea.

Condition	Effort	Calls	Parse rate	Input tokens	Output tokens	Total tokens	Words per idea
A · Full Graph	low	65	100%	10,912	3,331	14,243	91
A · Full Graph	medium	65	100%	10,912	5,295	16,208	99
A · Full Graph	high	65	100%	10,912	11,163	22,076	108
A · Full Graph	xhigh	30	100%	10,912	28,346	39,258	117
B · Slice	low	65	100%	3,778	2,849	6,627	89
B · Slice	medium	65	100%	3,778	4,772	8,550	98
B · Slice	high	65	98.5%	3,778	8,628	12,407	100
B · Slice	xhigh	30	100%	4,135	25,885	30,020	117
R · Random	high	65	100%	3,143	8,778	11,921	98

Table 8 (continued)

Condition	Effort	Calls	Parse rate	Input tokens	Output tokens	Total tokens	Words per idea
C · Control	high	65	100%	1,055	6,939	7,994	106

xhigh was run on 6 of the 13 market needs; those market needs have larger slices on average, which is why B's xhigh input is higher.

- **Input:** the slice cuts input tokens by 65% on average (slices range from 10 to 51 nodes, 8–41% of the graph). R matches B's node counts per layer but uses 17% fewer input tokens, because random nodes have shorter labels and fewer linking edges; the size match is close but not exact.
- **Total:** the slice saves 53% of total tokens at low effort, 47% at medium, 44% at high and 24% at xhigh. Output grows steeply with effort and soon dominates the bill, so the input saving matters less and less.
- **Output grows with context:** at high effort, output is 6.9k tokens with no graph, 8.6–8.8k with a slice (B or R) and 11.2k with the full graph. More context makes the model think and write more, a cost beyond the prompt itself.
- **Output by effort:** each step up costs more than the last: +1,900–2,000 tokens from low to medium, +3,900–5,900 from medium to high, about +17,000 from high to xhigh.
- **Validity:** 579 of 580 calls returned valid JSON (576 with 5 ideas, three with 3 or 6 ideas); none hit the token limit.

4.2.1 How the ideas use the graph

Table 9 shows where the ideas anchor in the graph and how much they add to it.

Table 9: How the ideas use the graph, by representation and effort: where ideas anchor, how often they repeat the prompt's worked example, and how many new nodes they propose.

Condition	Effort	Anchors outside the market need's own slice	Anchors to nodes not in the graph	Ideas repeating the worked example (of 5)	New nodes per idea
A · Full Graph	low	53%	0%	0.95	2.8
A · Full Graph	medium	49%	0%	0.69	3.1
A · Full Graph	high	48%	0%	0.54	3.4
A · Full Graph	xhigh	40%	0%	0.33	3.6
B · Slice	low	0%	0%	0.86	2.9
B · Slice	medium	0%	0%	0.48	3.1
B · Slice	high	0%	0%	0.55	3.3
B · Slice	xhigh	0%	0%	0.40	3.9
R · Random	high	42%	0%	0.51	3.3
C · Control	high	—	48%	0.37	4.4

- With the full graph, about half of all anchors lie outside the target market need's slice: A uses its extra context to reach across the product.
- With a random slice, 42% of anchors fall outside the market need's slice, because most of what R shows is unrelated to the market need.
- With no graph, the model invents node IDs for half of its anchors and adds more new nodes per idea (4.4): it builds its own picture of the product instead of working on the given one.
- Repetition of the worked example falls with effort, and is lowest with no graph.

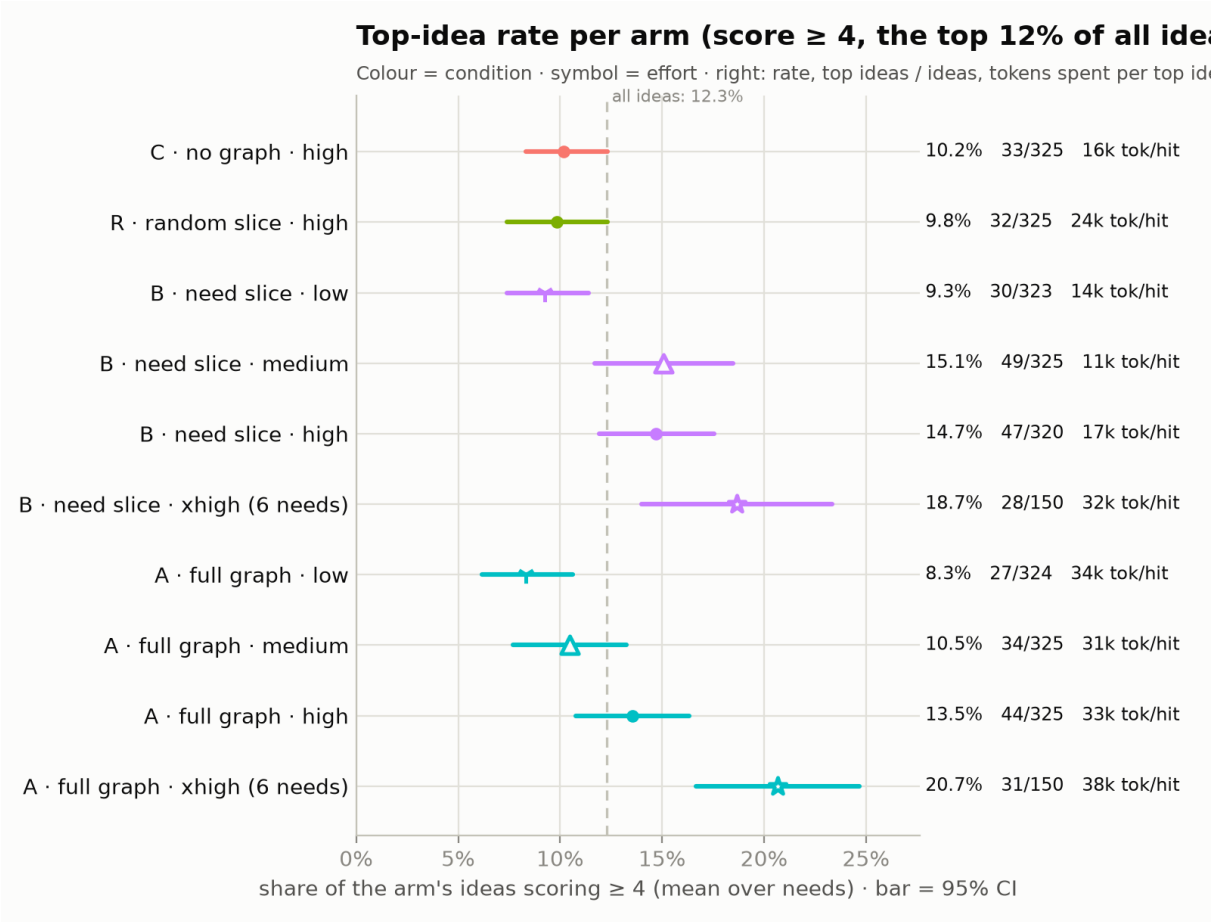


Figure 6: Share of each arm's ideas that are top ideas (score ≥ 4.0, the top 12% of all ideas), averaged over market needs, with 95% intervals from resampling calls within market need. This is quality only; cost per call is not taken into account here.

4.3 RQ1: Which representation yields the most top ideas per token?

Answer. The market-need slice. At medium effort it yields 8.8 top ideas per 100k tokens, about three times the full graph at any effort (2.6–3.2), and it is more efficient than the full graph on 10 of 13 market needs. At high effort, where all four representations were run, yields rank no graph (6.4) $\approx$ market-need slice (5.9) $>$ random slice (4.1) $>$ full graph (3.1).

A **top idea** scores ≥ 4.0 on the individual blind scoring. Scores move in 0.25 steps, so an exact top 10% does not exist: ≥ 4.0 keeps the top 12.3% of all ideas (355 of 2,892), ≥ 4.25 the top 2.4% (68). Market needs differ a lot in how easy a top idea is (29% of M14's ideas reach 4.0, 3% of M11's), so every rate is the mean of per-market-need rates.

4.3.1 Top-idea rate (score ≥ 4.0)

Figure 6 shows the rates with their intervals; Table 10 gives them with their cost.

Table 10: Top-idea rate per arm (share of ideas scoring ≥ 4.0), averaged over market needs with 95% bootstrap intervals, with cost per call and per top idea. Bold: lowest cost per top idea.

Arm	Needs	Top ideas / ideas	Top-idea rate	95% CI	Calls with ≥ 1 top idea	Tokens per call	Tokens per top idea
C · Control · high	13	33 / 325	10.2%	8.3–12.3%	48%	7,994	15,746
R · Random · high	13	32 / 325	9.8%	7.7–12.3%	45%	11,921	24,215
B · Slice · low	13	30 / 323	9.3%	7.4–11.4%	37%	6,627	14,359
B · Slice · medium	13	49 / 325	15.1%	11.7–18.5%	52%	8,550	**11,342**
B · Slice · high	13	47 / 320	14.7%	11.9–17.5%	56%	12,384	16,838
B · Slice · xhigh	6	28 / 150	18.7%	14.0–23.3%	73%	30,020	32,164
A · Full Graph · low	13	27 / 324	8.3%	6.2–10.5%	37%	14,243	34,289
A · Full Graph · medium	13	34 / 325	10.5%	8.0–13.2%	45%	16,208	30,985
A · Full Graph · high	13	44 / 325	13.5%	10.5–16.3%	57%	22,076	32,612
A · Full Graph · xhigh	6	31 / 150	20.7%	16.7–24.7%	80%	39,258	37,991

- **Full graph vs slice:** at every effort, B's top-idea rate is equal to or higher than A's, at about half the tokens per top idea.
- **Cheapest top ideas:** B at medium effort (11k tokens each), then B at low (14k), C (16k) and B at high (17k). A costs 31–38k at every effort.

4.3.2 Yield: top ideas per 100k tokens

The top-idea rate says how good an arm's ideas are; the yield says how fast it finds top ideas for the money. Counting top ideas against the tokens spent on an arm's calls (taken in replicate order, cycling through the market needs) gives a running count that grows in a straight line for every arm (Figure 7). The **yield** is the slope of that line: all of the arm's top ideas divided by all its tokens, per 100k tokens, with a 95% CI from 2,000 resamples of calls within each market need. Unlike a probability, a count does not level off, so arms are compared directly by slope. Table 11 gives the yields.

Table 11: Yield per arm, the main measure: top ideas per 100k tokens with 95% bootstrap intervals, ranked from most to least efficient.

Arm	Top ideas	Tokens spent	Yield (top ideas per 100k tokens)	95% CI
B · Slice · medium	49	0.56M	**8.8**	7.0–10.7
B · Slice · low	30	0.43M	7.0	5.4–8.4
C · Control · high	33	0.52M	6.4	5.1–7.7
B · Slice · high	47	0.79M	5.9	4.9–7.0
R · Random · high	32	0.77M	4.1	3.1–5.2
A · Full Graph · medium	34	1.05M	3.2	2.5–4.1
B · Slice · xhigh (6 market needs)	28	0.90M	3.1	2.4–3.9
A · Full Graph · high	44	1.43M	3.1	2.4–3.7

Table 11 (continued)

Arm	Top ideas	Tokens spent	Yield (top ideas per 100k tokens)	95% CI
A · Full Graph · low	27	0.93M	2.9	2.2–3.7
A · Full Graph · xhigh (6 market needs)	31	1.18M	2.6	2.1–3.1

- **The market-need slice is the most efficient representation at every effort up to high.** B at medium yields 8.8 top ideas per 100k tokens, 2.8 times A at medium (3.2); at high, 5.9 against 3.1. The intervals do not overlap.
- **How far apart the slopes are** (Figure 8, chance that each arm's slope beats the next lower arm's): B · medium over B · low 0.93; B · low over C 0.74; C over B · high 0.69; B · high over R 0.99; R over A · medium 0.91; among the full-graph arms and B · xhigh 0.53–0.73. Four tiers: B at medium (8.8); B at low and high with C (5.9–7.0); R (4.1); all full-graph arms and B · xhigh (2.6–3.2).
- **No graph:** C's low top-idea rate (10.2%) is offset by its cheap calls (8.0k tokens), giving a yield of 6.4, above R and A and within the interval of B at high effort. Its weakness is coverage: with no graph, M1 produced no top idea in 5 calls (A and B: 2–3).
- **At the stricter threshold (≥ 4.25)** the order is similar but the counts are small (2–11 per arm): B · medium 1.8 per 100k tokens (1.1–2.7), A · medium 1.0, B · low 0.9, B · high 0.9, A · xhigh 0.85, B · xhigh 0.8, C 0.8, A · high 0.7, R 0.4, A · low 0.2.
- **On the 6 xhigh market needs only:** B · medium 9.0, B · low 7.4, B · high 6.4, C 6.1, R 4.2, A · low 3.3, B · xhigh 3.1, A · medium 2.9, A · high 2.8, A · xhigh 2.6. The ranking is the same as on all market needs.

4.3.3 Yield per market need

To check that the yield ranking is not driven by a few easy market needs, yields were computed for each market need separately (5 calls per arm and market need), and arms were compared by the number of needs on which each is more efficient (exact two-sided sign test, ties excluded; `yield_per_market_need.xlsx`).

Table 12 gives the yield per market need at high effort (top ideas per 100k tokens; B's slice size in nodes), and Table 13 the sign tests.

Table 12: Yield per market need at high effort (top ideas per 100k tokens), ordered by the size of the market-need slice. The last column is the slice's yield divided by the full graph's.

Need	Slice nodes	B market-need slice	C no graph	R random slice	A full graph	$B \div A$
M11	10	2.1	0.0	2.1	1.7	1.2×
M14	15	17.9	15.5	10.6	8.2	2.2×
M16	15	9.2	5.1	2.5	3.0	3.1×
M8	25	13.4	10.1	7.6	2.9	4.7×
M6	28	11.8	7.4	5.8	3.4	3.5×
M7	45	5.5	12.8	5.8	2.8	2.0×
M13	45	6.0	5.0	3.1	2.7	2.2×
M3	47	4.1	10.0	4.8	4.5	0.9×
M12	47	3.1	2.6	1.6	2.8	1.1×
M1	48	2.6	0.0	0.0	3.2	0.8×
M4	51	4.3	6.8	4.1	1.9	2.3×
M5	51	0.0	0.0	1.5	0.0	—
M17	51	2.9	7.9	6.1	2.8	1.0×

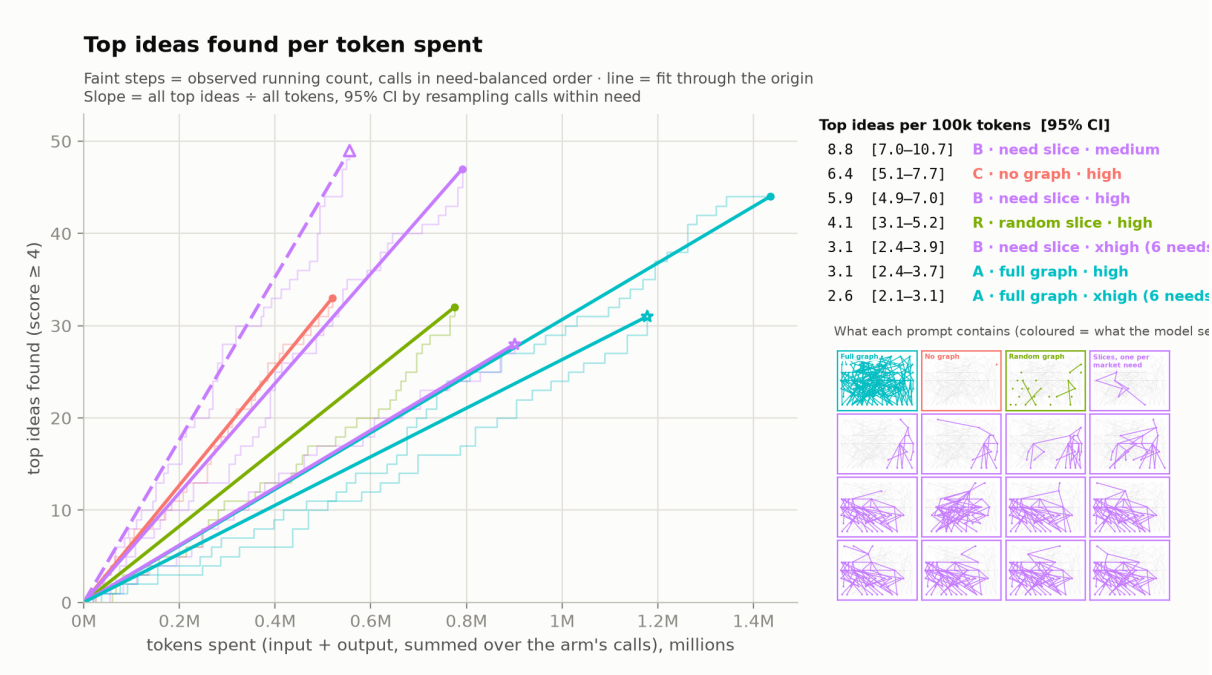


Figure 7: Top ideas (score $\geq$ 4.0) found against tokens spent, per arm, with calls taken in replicate order. Each arm accumulates top ideas at a near-constant rate, so its slope is its yield (top ideas per 100k tokens): the steeper the line, the more efficient the arm. The market-need slice at medium effort climbs fastest; the full-graph arms are the flattest at every effort.

Table 13: Arm-versus-arm comparisons across market needs: number of needs on which each arm has the higher yield, with exact two-sided sign tests (ties excluded). Bold: $p < 0.05$.

Comparison	First arm more efficient	Second more efficient	Equal	Sign test p
B · Slice vs A · Full Graph, high effort	10 market needs	2	1	**0.039**
B · Slice vs A · Full Graph, medium effort	10	3	0	0.092
C · Control vs R · Random	10	2	1	**0.039**
R · Random vs A · Full Graph	10	3	0	0.092
C · Control vs A · Full Graph	9	3	1	0.146
B · Slice vs R · Random	9	4	0	0.267
B · Slice vs C · Control	8	4	1	0.388

- **The slice's efficiency advantage over the full graph holds market need by market need:** B is more efficient on 10 of 13 market needs at high effort (p = 0.039) and at medium effort (p = 0.09). The full graph is ahead only on M1 and M3, by small margins.
- **Single needs are noisy:** each value rests on 5 calls and 0–10 top ideas.

4.3.4 Chance of at least one top idea for a given budget (secondary)

Figure 9 shows the curves.

This measure answers a practical question, how likely one budget is to produce at least one top idea, but every arm's chance approaches 100% as the budget grows, so it compresses differences at large budgets and can suggest that spending alone guarantees a top idea. The yield above is the better basis for comparing arms. For a token budget spent on whole calls of one arm, the chance that at least one idea scores $\geq$ 4.0, averaged over market needs. Within a market need, calls are treated as independent: k calls succeed with $1 - (1 - p)^k$, where p is the

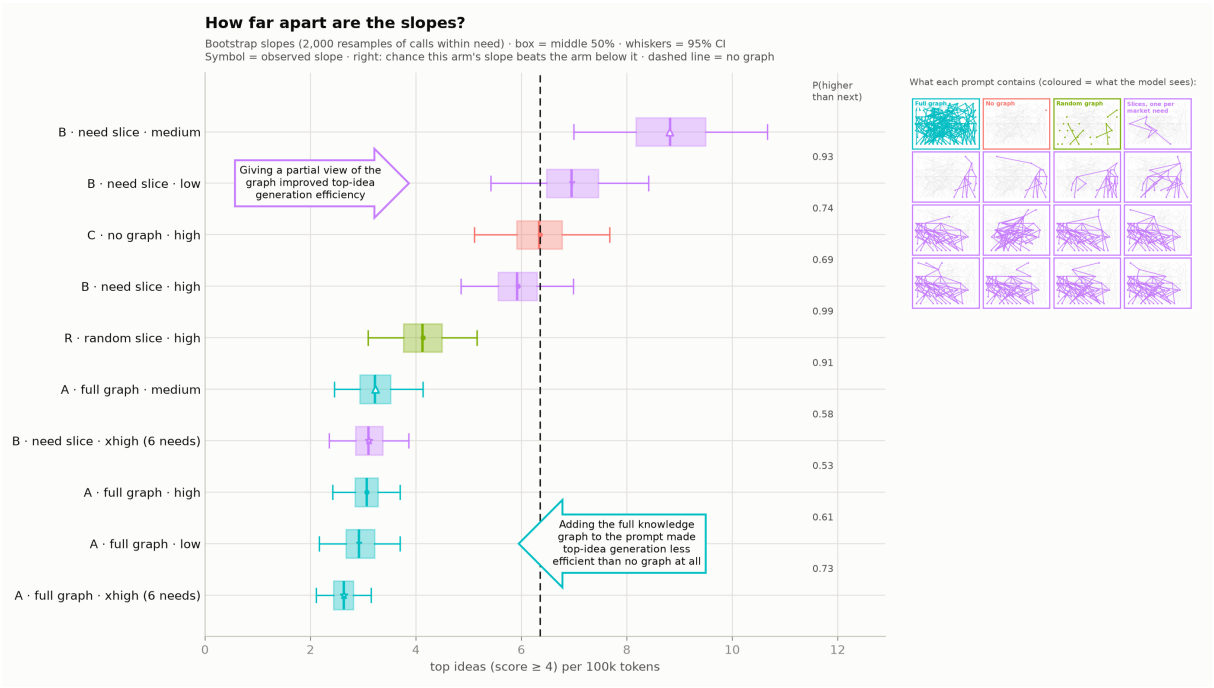


Figure 8: How certain the yields are, *given the judge's scores*: for each arm, a box plot of 2,000 bootstrap slopes (calls resampled within market need; box = middle 50%, whiskers = 95% interval). The intervals capture variation between generation calls only. They do not include the variability of the judge's scores on repeated evaluation, and the 4.0 threshold was chosen after seeing the Experiment 1 scores (Section 3.7). The dashed line marks the control, no knowledge graph in the prompt (C, high effort). Arms whose boxes lie to the left of the line find top ideas less efficiently than giving the model no graph at all; this includes every full-graph arm, the random slice of the graph, and the slice of graph dangling from a given market-need slice. When using a graph slices, only setting at low and medium effort underperform the control "C"

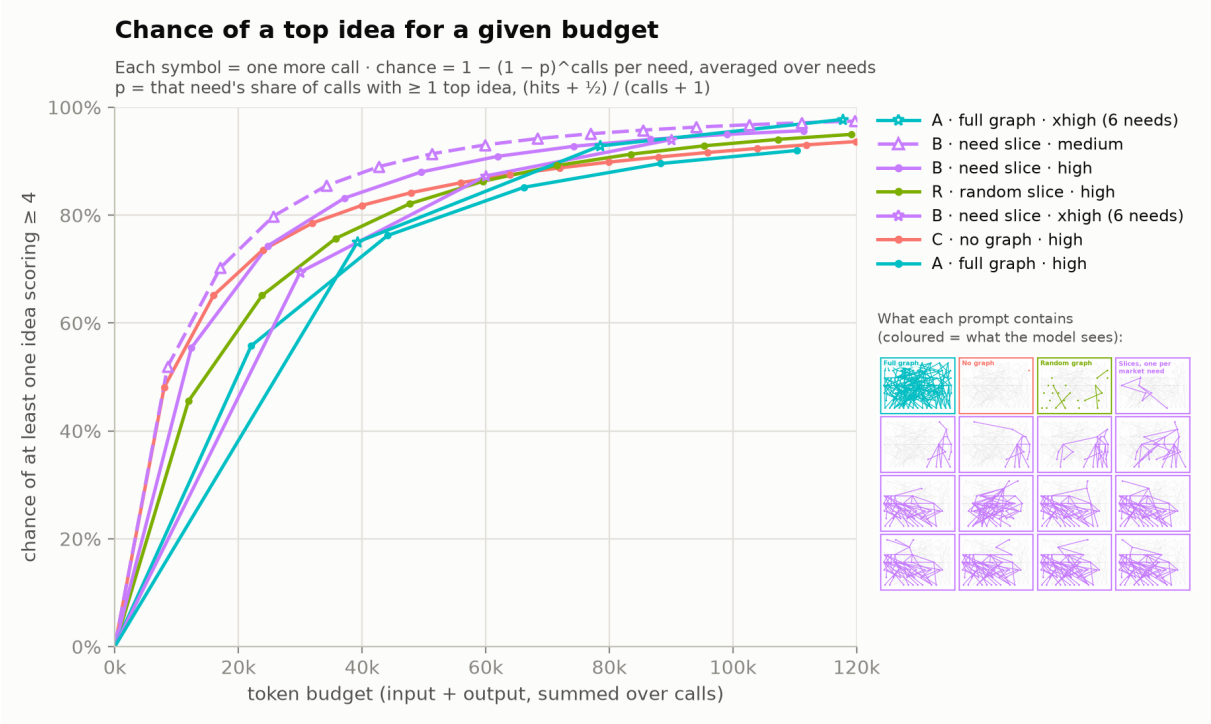


Figure 9: Chance of at least one top idea for a given token budget, per arm, from whole calls and a Jeffreys estimate of each market need's success rate. Every curve rises towards 1 as the budget grows, so this view compresses differences at large budgets; yield (Figure 7) is the main measure.

market need's share of calls with a top idea, estimated as $(\text{hits} + \frac{1}{2})/(\text{calls} + 1)$ because 5 calls per market need often give exactly 0 or 1.

Table 14 gives the chance at three budgets.

Table 14: Chance of at least one top idea for a given token budget, per arm. In brackets: the number of whole calls that budget buys.

Arm	25k tokens	50k tokens	100k tokens
C · Control · high	74% (3 calls)	84% (6)	92% (12)
R · Random · high	65% (2)	82% (4)	93% (8)
B · Slice · low	64% (3)	81% (7)	91% (15)
B · Slice · medium	70% (2)	**89%** (5)	**96%** (11)
B · Slice · high	74% (2)	88% (4)	95% (8)
B · Slice · xhigh (6 market needs)	— (0)	69% (1)	94% (3)
A · Full Graph · low	39% (1)	67% (3)	86% (7)
A · Full Graph · medium	46% (1)	79% (3)	94% (6)
A · Full Graph · high	56% (1)	76% (2)	90% (4)
A · Full Graph · xhigh (6 market needs)	— (0)	75% (1)	93% (2)

- **B at medium and high effort lead at 50k and 100k tokens,** and B-medium is at or near the top at every budget.
- **No graph is a good choice for small budgets** because its calls are cheapest, but it rises more slowly at large budgets because of the market needs where it finds nothing (M1); M5 and M11 are hard for every arm.
- **The full graph trails at every budget:** its calls are expensive and its top-idea rate is no higher than B's.

4.3.5 Stricter threshold and matched market needs

- **Score ≥ 4.25 (top 2.4%):** same direction, too few hits per arm (2–11) to separate arms. A · xhigh 6.7% and B · xhigh 4.7% (6 market needs), A · medium 3.4%, B · medium 3.1%, A · high 3.1%, B · high 2.2%, C 1.2%, B · low 1.2%, R 1.0%, A · low 0.6%. Tokens per top idea are lowest for B · medium (56k) and highest for R (about 260k) and A · low (463k); yields are listed in the yield section above.
- **Only the 6 xhigh market needs** (`--match-needs`): B · medium and B · high 16.7%, B · xhigh 18.7%, A · high 12.7%, A · xhigh 20.7%, R 10.7%, C 10.0%. On equal market needs, B gains little from xhigh over medium or high, while A gains a lot; A needs the extra thinking to make use of its larger context.

4.3.6 The top ideas

The 10 highest-scoring of all 2,892 ideas (`results/top_ideas.md`), all scoring 4.50: 5 from A (high ×2, medium ×2, xhigh), 4 from B (high ×2, low, xhigh) and 1 from R (high). The bottom 10 are dominated by B at low effort (6 of 10).

- **The best ideas converge on two themes from many arms:** replacing the magnetron and high-voltage chain with a solid-state GaN RF source (for M13 energy efficiency, and from R for M7 safety), and solid-state 27 MHz or 915 MHz RF thawing for M4. Arms differ in how often they reach these ideas, not in whether they can.
- **The same technical theme scores low when it does not serve the target market need** (e.g. a GaN power stage proposed for M12, healthy diet): the judge rewards fit to the market need, not technical appeal alone.
- Many ideas tie at the top, so the ranked list identifies top-scoring ideas rather than a strict order.

4.4 RQ2: Does the amount or the relevance of the graph matter?

Answer. Relevance, not amount. More graph buys better average sets: the full graph wins 64–65% of set comparisons against the market-need slice, mainly on specificity, but its top-idea rate is no higher, so at about twice the tokens per call it finds fewer top ideas per token. Relevant structure helps in the top tail: the market-need slice out-yields a random slice of the same size in 99% of paired bootstrap draws, although whole-set judging cannot tell the two apart. For Haiku, irrelevant structure is worse than none: no graph is more efficient than the random slice on 10 of 12 market needs. Stronger models are less affected: in Experiment 2 the random slice was about as efficient as no graph for Opus, and between the full graph and no graph for Fable (Section 4.6).

4.4.1 Amount: full graph vs market-need slice

Figure 10 shows every pairwise set comparison in this section and in Section 4.5; Figure 11 shows how the full graph and the slice differ on each criterion. Table 15 gives the win rates and Table 16 the differences per criterion.

Table 15: Pairwise set judging, full graph versus market-need slice at equal effort: wins, win rate with 95% Wilson interval, sign test, and agreement between the two presentation orders.

Effort	Pairs	Full graph wins	Slice wins	Ties	Full graph win rate	95% CI	Sign test p	Order agreement
low	65	37	18	10	65%	53–75%	0.015	85%

Table 15 (continued)

Effort	Pairs	Full graph wins	Slice wins	Ties	Full graph win rate	95% CI	Sign test p	Order agreement
high	65	34	16	15	64%	52–74%	0.015	77%

Mean score difference, Full graph minus Slice (1–5 scale):

Table 16: Full graph minus market-need slice: mean difference in set scores per criterion (1–5 scale), by effort.

Effort	Relevance	Novelty	Feasibility	Specificity	Diversity
low	+0.13	−0.01	+0.33	+0.27	+0.17
high	−0.01	+0.06	+0.09	+0.34	+0.16

- **The full graph yields better sets:** A wins 65% of set comparisons at low effort and 64% at high (p = 0.015 at both).
- The full graph's advantage on whole sets is real but modest, mainly in specificity; novelty and relevance are about equal.
- **The slice is not shown to be non-inferior on sets:** B's win rate is 35–36%, with the lower confidence bound (25–26%) well below the 45% non-inferiority margin. On the tail measure, however, B is not worse than A at any effort.
- **The effect varies by market need:** A swept 5–0 for M17 and M6 at high effort and M11 at low; B led for M1 at low effort, M3 at both levels and M8 at high. With 5 pairs per market need, no single market need is significant.

4.4.2 Relevance in the top tail: market-need slice, random slice and no graph

The market-need slice (B) and the random slice (R) have the same number of nodes per layer, so comparing them separates relevance from amount; no graph (C) is the floor.

- **Top-idea rate, B vs R:** at the same effort and nearly the same size, the market-need slice yields top ideas at 14.7% against 9.8% for the random slice; the confidence intervals just touch. The whole-set judging does not confirm this (Section 4.4.3).
- **Yield, B vs R:** at high effort, B yields 5.9 and R 4.1; the 95% intervals just touch (4.9–7.0 against 3.1–5.2), but pairing the two arms' bootstrap draws, B's slope is the higher in 99% of them (Figure 8), equivalent to a one-sided bootstrap $p \approx 0.01$. R costs about as much per call as B, so the difference is in what the calls find.
- **Per market need, B vs R is not robust** (9 of 13, p = 0.27), in line with the pooled yield (intervals touching) and the set judging (no difference).
- **Graph vs none:** a relevant slice yields more top ideas than no graph (14.7% vs 10.2%); a random slice does not (9.8%). The full graph is in between (13.5%).
- **No graph is more efficient than a random slice on 10 of 12 decided market needs (p = 0.039).** Showing the model structure that is not about the problem costs more per top idea than showing it nothing.

4.4.3 Pairwise set judging: the controls (high effort)

Table 17 gives the win rates against the controls and Table 18 the differences per criterion.

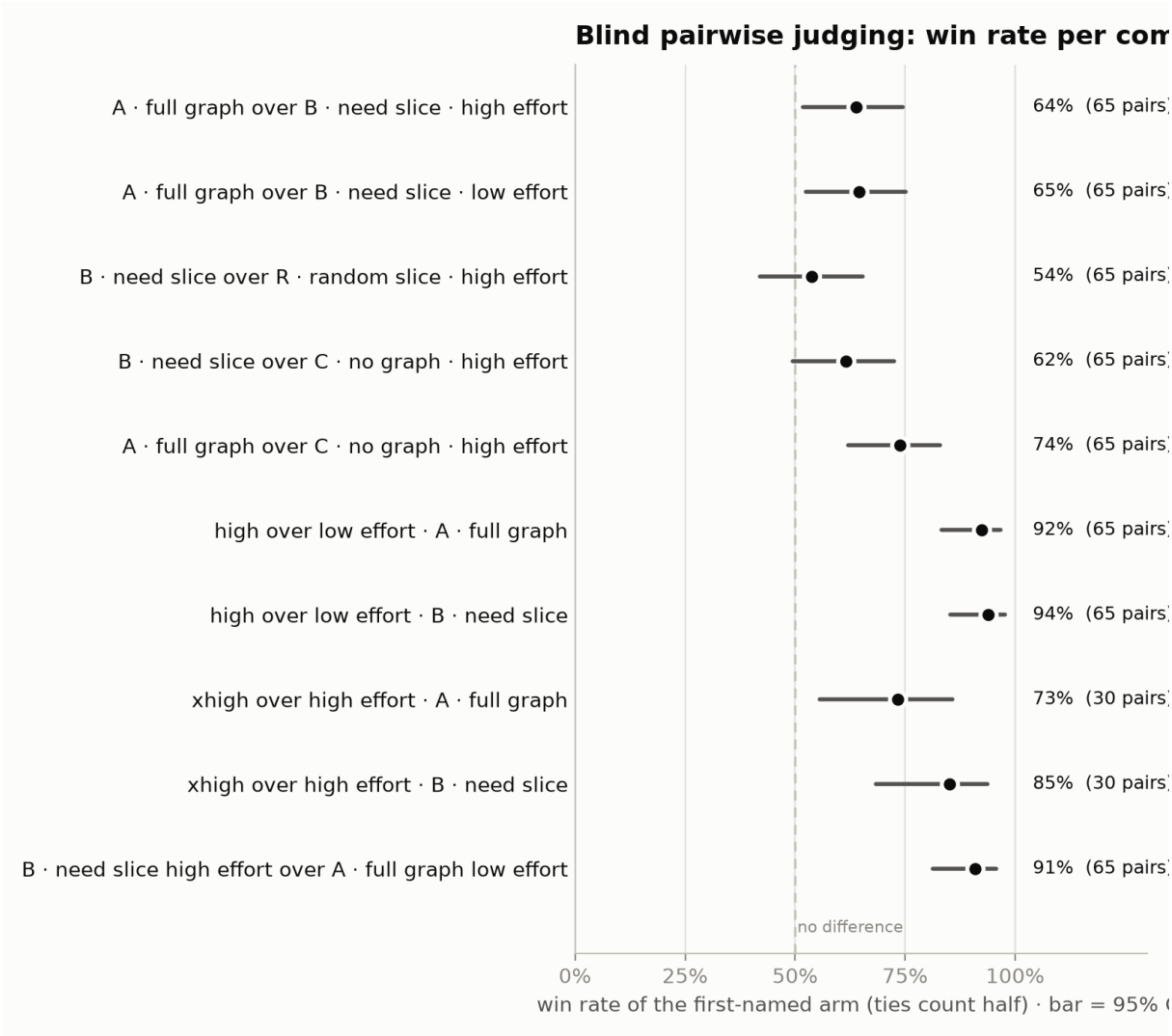


Figure 10: Every pairwise set comparison: the first arm's win rate with a 95% Wilson interval (each pair judged in both presentation orders, ties excluded). Values above 50% favour the first arm.

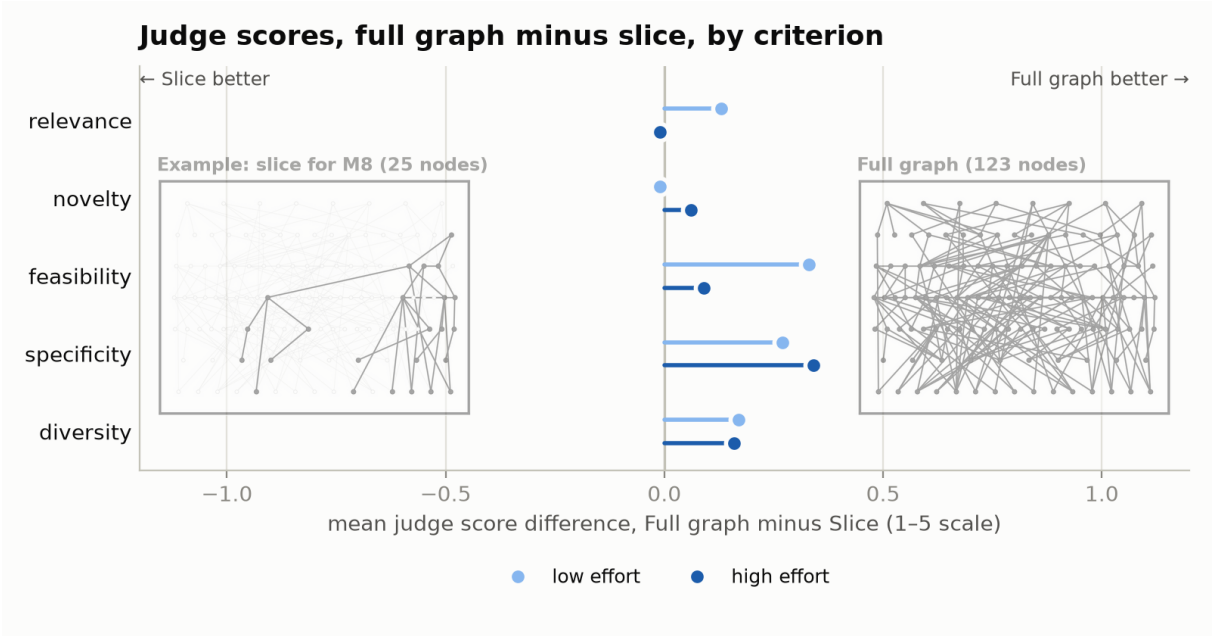


Figure 11: Full graph minus market-need slice (A − B) on each set-judging criterion, by effort. The full graph's advantage is mainly specificity. A human expert reviewing these criteria cautions that they may partly reflect how the ideas are worded rather than what they propose.

Table 17: Pairwise set judging against the controls (no graph and random slice) at high effort.

Comparison	Pairs	First wins	Second wins	Ties	First-arm win rate	95% CI	Sign test p	Order agreement
A · Full Graph vs C · Control	65	44	13	8	74%	62–83%	< 0.001	88%
B · Slice vs C · Control	65	34	19	12	62%	49–72%	0.053	82%
B · Slice vs R · Random	65	25	20	20	54%	42–65%	0.55	68%
A · Full Graph vs B · Slice, for reference	65	34	16	15	64%	52–74%	0.015	77%

Mean score difference, first minus second arm (1–5 scale):

Table 18: Comparisons with the controls: mean difference in set scores per criterion, first minus second arm (1–5 scale).

Comparison	Relevance	Novelty	Feasibility	Specificity	Diversity
A · Full Graph − C · Control	+0.21	+0.24	+0.29	+0.88	−0.11
B · Slice − C · Control	+0.16	+0.20	+0.14	+0.52	−0.30
B · Slice − R · Random	+0.14	−0.08	+0.20	−0.03	−0.20

- **Full graph vs no graph:** a clear win for A on every criterion except diversity, above all specificity (+0.88). R vs C was not judged.
- **Need slice vs no graph:** B wins 62%, just short of significance (p = 0.053), again mainly

through specificity. C's sets are judged the more diverse (−0.30): without a graph, the five ideas range more widely.

- **Need slice vs random slice:** no detectable difference. B's sets are slightly more relevant and feasible, R's slightly more diverse; 20 of 65 pairs are ties and the judge agreed with itself across orders for only 68% of pairs, the lowest of any comparison, so the two kinds of set are hard to tell apart.
- **Together with the tail:** the graph's set-level benefit grows with the amount of real product structure in the prompt (C < B < A). Relevance shows up mainly in the top tail (B 14.7% vs R 9.8% top ideas), not in the average set. This matters for the claim: a relevant slice helps find the best ideas, but the evidence that it beats a random slice of the same size comes from the tail measure alone.
- **Length does not explain the wins over no graph:** C's sets are longer than B's (529 vs 492 words), yet B wins; in A vs C the longer set won 68% of decided pairs, below A's 74%.

4.4.4 Mean idea quality (individual scoring)

Table 19 gives the need-balanced mean idea score (1–5) and mean criterion scores.

Table 19: Mean idea score and mean criterion scores per arm (individual scoring, 1–5, balanced across market needs).

Arm	Mean score	Relevance	Novelty	Feasibility	Specificity
C · Control · high	3.44	3.49	3.29	3.21	3.78
R · Random · high	3.46	3.44	3.16	3.37	3.88
B · Slice · low	3.28	3.38	3.01	3.26	3.50
B · Slice · medium	3.47	3.53	3.20	3.36	3.82
B · Slice · high	3.53	3.60	3.25	3.37	3.92
B · Slice · xhigh (6 market needs)	3.54	3.47	3.38	3.29	4.03
A · Full Graph · low	3.30	3.39	3.04	3.27	3.52
A · Full Graph · medium	3.44	3.42	3.10	3.42	3.83
A · Full Graph · high	3.50	3.46	3.20	3.45	3.89
A · Full Graph · xhigh (6 market needs)	3.52	3.42	3.33	3.29	4.03

- Mean differences between arms are small (0.1–0.25 points) and hide the differences in the tail; this is why the tail is the main measure.
- At high effort, mean scores order C 3.44 < R 3.46 < A 3.50 < B 3.53, the same ranking as the top-idea rates except that C and R, nearly tied on both measures, swap places.
- **No graph gives the most novel but least feasible single ideas** at high effort: novelty 3.29 (highest at high effort) and feasibility 3.21 (lowest). Without the graph, ideas are less constrained by the actual product. Judged as sets, though, A and B are rated more novel than C (Section 4.4.3): the individual novelty advantage of C does not survive a side-by-side comparison.
- **Individual ideas vs whole sets:** B's ideas score at least as well as A's individually, yet A wins 64% of set comparisons. A's advantage is at the level of the set (specificity, diversity), not in better single ideas.

4.5 RQ3: How does spending tokens on context compare with spending them on thinking effort?

Answer. Thinking effort raises quality; the choice of context raises efficiency. High effort beats low in 92–94% of set comparisons and xhigh gives the highest top-idea rates (19–21%), but the cost per call rises in step, so yield does not improve: it is flat for the full graph and peaks at medium effort for the slice. For a similar budget per call, tokens are better spent on thinking

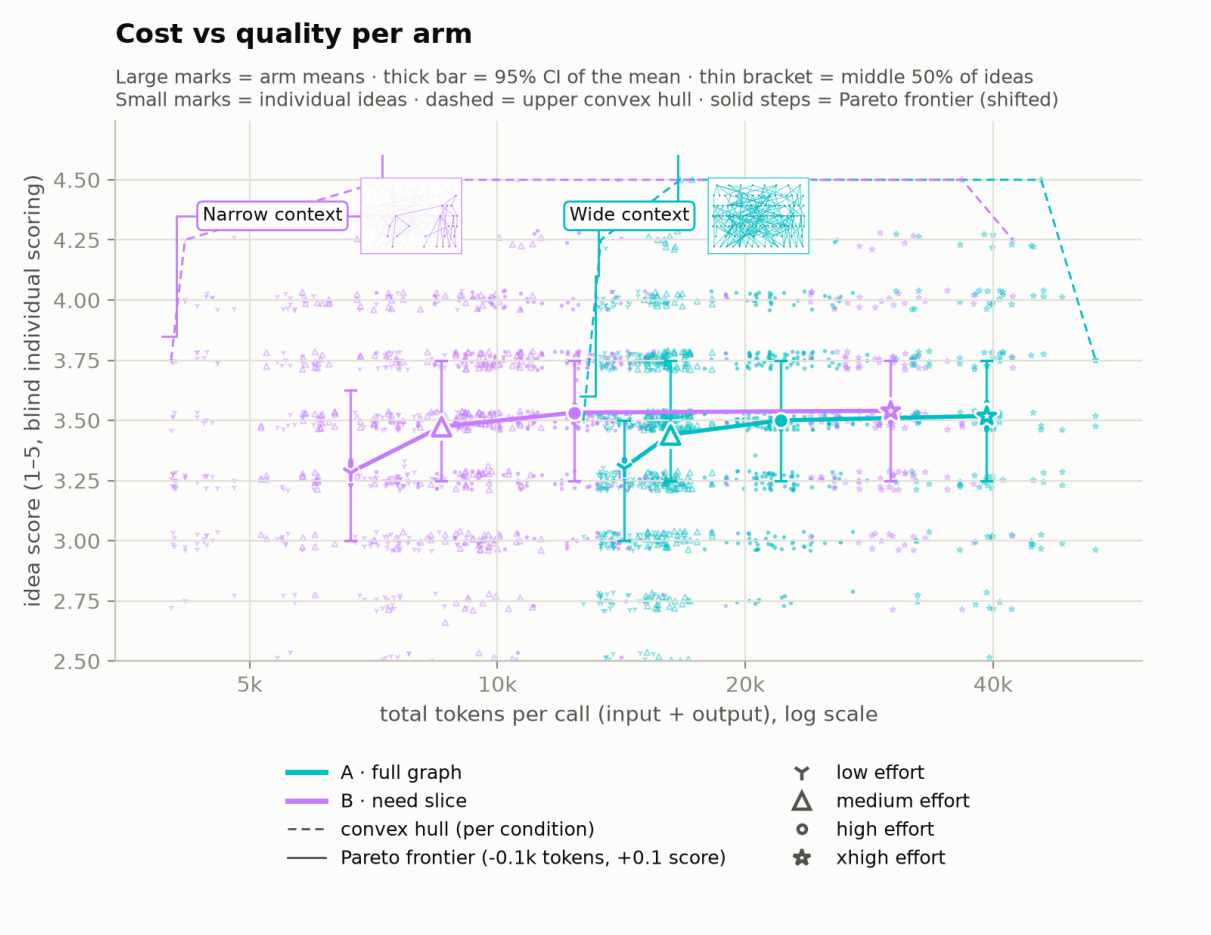


Figure 12: Tokens per call against mean idea score for the full-graph (A) and market-need-slice (B) arms at each effort. Moving right costs tokens; moving up raises the mean idea score (the main measure, yield, is reported separately).

with a relevant slice than on context: the slice at high effort beats the full graph at low effort in 91% of set comparisons, with 13% fewer tokens per call.

4.5.1 Effort, quality and yield

Figure 12 plots tokens per call against mean idea score for the full-graph and market-need-slice arms.

- **Effort:** top-idea rates rise with effort for both A and B, with the largest rates at xhigh. But xhigh costs 2–3× the tokens per top idea of medium or high.
- **The full graph's yield is flat across effort** (2.6–3.2). More thinking raises A's top-idea rate from 8% to 21%, but the cost per call rises in step, so the full graph buys better calls, not more top ideas per token.
- **The slice's yield peaks at medium effort.** From medium to high, B's top-idea rate barely changes (15.1% to 14.7%) while the tokens per call rise by 45%; xhigh raises the rate to 18.7% but at 3.5 times the tokens.

4.5.2 Pairwise set judging: effort levels (equal representation)

Table 20 gives the win rates between effort levels and Table 21 the differences per criterion.

Table 20: Pairwise set judging between effort levels within the same representation.

Comparison	Condition	Pairs	Higher wins	Lower wins	Ties	Higher-effort win rate	95% CI	Sign test p	Order agreement
high vs low	A · Full Graph	65	58	3	4	92%	83–97%	< 0.001	94%
high vs low	B · Slice	65	59	2	4	94%	85–98%	< 0.001	94%
xhigh vs high	A · Full Graph	30	19	5	6	73%	56–86%	0.007	80%
xhigh vs high	B · Slice	30	25	4	1	85%	68–94%	< 0.001	97%

Mean score difference, higher minus lower effort:

Table 21: Effort comparisons: mean difference in set scores per criterion, higher minus lower effort (1–5 scale).

Comparison	Condition	Relevance	Novelty	Feasibility	Specificity	Diversity
high − low	A · Full Graph	+0.48	+0.71	+0.52	+1.14	+0.63
high − low	B · Slice	+0.71	+0.69	+0.70	+1.23	+0.71
xhigh − high	A · Full Graph	+0.25	+0.34	+0.02	+0.46	+0.33
xhigh − high	B · Slice	+0.19	+0.47	+0.03	+0.61	+0.45

- High beats low on every criterion and on all 13 market needs.
- xhigh beats high again, mainly through specificity, novelty and diversity, not feasibility, at 1.8× (A) to 2.4× (B) the tokens. Effort had not saturated at the highest level tested.

4.5.3 Equal budget: slice with high effort vs full graph with low effort

Table 22 compares two arms of similar cost per call.

Table 22: Market-need slice at high effort versus full graph at low effort, two arms of similar cost per call.

Comparison	Pairs	Slice-high wins	Full-graph-low wins	Ties	Slice-high win rate	95% CI	Sign test p	Order agreement
B · Slice · high vs A · Full Graph · low	65	56	3	6	91%	81–96%	< 0.001	91%

B-high was ahead on every criterion and for all 13 market needs, at 12,407 vs 14,243 tokens per call (13% fewer). For a similar budget, a relevant slice with more thinking clearly beats the full graph with little thinking.

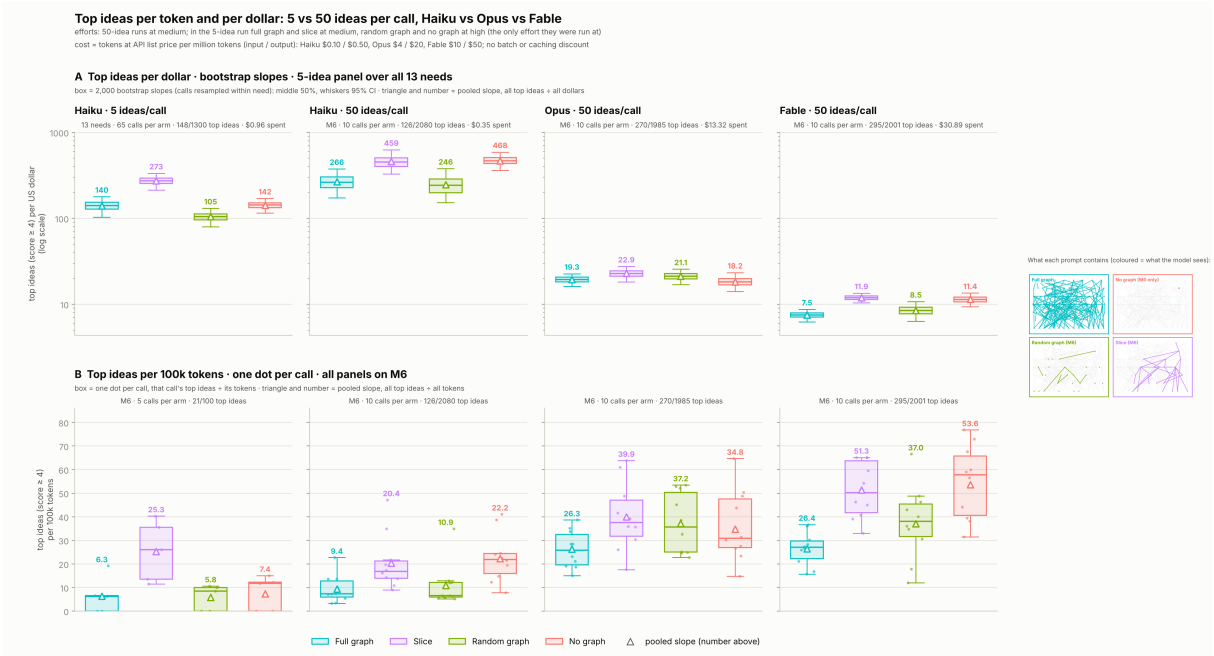


Figure 13: Yield by representation and model. Columns: Haiku with 5 ideas per call (Experiment 1), then Haiku, Opus and Fable with 50 ideas per call on M6 (Experiment 2). **(A)** Top ideas per US dollar at list price (log scale). Each box holds 2,000 bootstrap estimates of the arm's pooled yield: box = middle 50%, whiskers = 95% CI; it shows how uncertain the arm's average is. The 5-idea column pools all 13 market needs. **(B)** Top ideas per 100k tokens, M6 only. Each dot is one call's yield (its top ideas divided by its tokens), so the box shows how much single calls vary (whiskers 1.5 × IQR). In both rows, the triangle and the number give the pooled yield, all top ideas divided by all cost. Experiment 2's ideas are shorter and score lower than Experiment 1's, so compare arms within a column, not heights across the first and the other columns.

4.6 RQ4: Does the effect of the representation hold across models, and what does each model cost per top idea?

Experiment 2 (Section 3.9) asked Haiku, Opus and Fable for 50 ideas per call on market need M6, ten calls per representation. Figure 13 shows the yield of every arm per dollar (A) and per token (B) next to Experiment 1's Haiku arms; Table 23 gives the values with 95% intervals.

Table 23: Experiment 2: top-idea rate and yield per arm (M6, 50 ideas per call, medium effort, 10 calls per arm), with bootstrap 95% intervals. Cost at list price.

Model	Representation	Top ideas / ideas	Per 100k tokens	Per US dollar
Haiku 5.5	A · Full Graph	28/517 (5.4%)	9.4 [6.3–13.2]	266 [177–373]
	B · Slice	42/525 (8.0%)	20.4 [14.5–27.8]	459 [326–630]
	R · Random	19/513 (3.7%)	10.9 [6.7–16.5]	246 [151–375]
	C · Control	37/525 (7.0%)	22.2 [17.0–28.5]	468 [360–600]
Opus 5.5	A · Full Graph	72/506 (14.2%)	26.3 [21.5–31.1]	19.3 [16.0–22.5]
	B · Slice	73/495 (14.7%)	39.9 [31.7–48.6]	22.9 [18.3–27.6]
	R · Random	60/487 (12.3%)	37.2 [29.7–45.6]	21.1 [16.8–25.5]
	C · Control	65/497 (13.1%)	34.8 [26.7–43.2]	18.2 [14.1–22.9]
Fable 5.1	A · Full Graph	78/501 (15.6%)	26.4 [22.0–30.6]	7.5 [6.2–8.8]
	B · Slice	87/507 (17.2%)	51.3 [45.0–57.9]	11.9 [10.4–13.5]
	R · Random	58/498 (11.6%)	37.0 [28.4–46.6]	8.5 [6.5–10.7]
	C · Control	72/495 (14.5%)	53.6 [44.3–63.4]	11.4 [9.4–13.6]

Table 23 (continued)

Model	Representation	Top ideas / ideas	Per 100k tokens	Per US dollar

4.6.1 Representation across models

The full graph is the least efficient representation for every model. Per token, its yield is about half that of the slice and of no graph for Haiku and Fable, and the lowest of the four for Opus. Its interval lies wholly below no graph's for Haiku and Fable and below the slice's for Fable; for Opus it only just separates from the slice's (upper bound 31.1 against lower bound 31.7).

The slice does no better than no graph. For all three models their intervals overlap almost completely (Haiku 20.4 and 22.2, Opus 39.9 and 34.8, Fable 51.3 and 53.6). Giving only the relevant slice removes the full graph's penalty but adds no gain over giving no graph at all.

For the stronger models, the full graph costs tokens, not quality. Opus and Fable produced top ideas at about the same rate with the full graph as with no graph (14.2% and 15.6% against 13.1% and 14.5%); their lower yield comes from the roughly 10,000 extra input tokens per call. For Haiku the full graph also lowered the rate (5.4% against 7.0%), as in Experiment 1.

The random slice was as poor as the full graph for Haiku but close to the slice and no graph for Opus, and in between for Fable. A stronger model seems less hurt by irrelevant context of moderate size.

4.6.2 Unit economics

Per token, the stronger models were about 1.5 to 2.5 times as productive (Opus 1.6–2.0×, Fable 2.4–2.5×). With the slice or no graph, Opus found 35–40 and Fable 51–54 top ideas per 100k tokens, against 20–22 for Haiku. All three used similar numbers of tokens per call (Table 7); the difference is the share of ideas that reach the top: 12–17% for Opus and Fable against 4–8% for Haiku.

Per dollar, the order reverses. At list prices Opus costs 40 times and Fable 100 times as much per token as Haiku, so Haiku found about 460 top ideas per dollar with the slice or no graph, against about 20 for Opus and 12 for Fable. A top idea cost about \$0.002 with Haiku, \$0.05 with Opus and \$0.09 with Fable. The whole run of 40 calls cost \$0.35 with Haiku, \$13.32 with Opus and \$30.89 with Fable.

Per dollar, the full graph's penalty shrinks. The graph adds only input tokens, which cost a fifth as much as output tokens. Per dollar, Opus's full graph is within the intervals of the other three representations (19.3 against 18.2–22.9); Fable's remains clearly below the slice and no graph (7.5 against 11.9 and 11.4).

The net cost depends on who sifts the ideas. The dollar yield counts only generation. But the top ideas must still be found among all the others, and Haiku makes the sifting longer: it produced one top idea per 16.5 ideas, against one per 7.4 for Opus and 6.8 for Fable. With a screening cost s per idea, the net cost of a top idea is

$$\text{net cost per top idea} = \frac{\text{generation cost} + s \times \text{ideas generated}}{\text{top ideas}}.$$

Table 24 evaluates it for the three models, pooled over the four representations, with 95% intervals from 2,000 bootstrap resamples of calls within representation. Screening by an automated judge, as in this study, cost \$0.0012 per idea (`claude-sonnet-5-5` at list price), and at that cost Haiku is the cheapest per top idea in every resample. Opus becomes cheaper than Haiku once screening costs more than about \$0.005 per idea (95% CI \$0.004–0.008), and Fable once it

costs more than about $0.01 ($0.008–0.015). At the cost of a person reading each idea (about 30 seconds at $60 an hour, $0.50 per idea), both larger models cost less than half as much per top idea as Haiku in every resample; between them, Opus and Fable cost about the same at any screening cost (Fable cheaper in 83% of resamples at $0.50). The content-only re-rating (Section 4.6.4) adds that Haiku's top ideas are also somewhat weaker. The cheapest model is therefore the one that matches the screening: a small model when an automated judge filters the output, a larger one when people do.

Table 24: Net cost per top idea (US dollars) for a given screening cost per idea, pooled over the four representations of Experiment 2, with bootstrap 95% intervals (2,000 resamples of calls within representation). Generation at list prices; $0.0012 is the measured cost of the automated judge per idea; $0.50 corresponds to about 30 seconds of human reading at $60 an hour. Bold: cheapest, or indistinguishable from the cheapest.

Model	Ideas per top idea	$0	$0.0012 (AI judge)	$0.50 (human)
Haiku 5.5	16.5 [14.1–19.5]	**0.003** [0.002–0.003]	**0.023** [0.019–0.027]	8.26 [7.07–9.78]
Opus 5.5	7.4 [6.7–8.2]	0.049 [0.045–0.055]	0.058 [0.053–0.065]	**3.73** [3.39–4.15]
Fable 5.1	6.8 [6.3–7.4]	0.105 [0.096–0.115]	0.113 [0.104–0.124]	**3.50** [3.24–3.82]

4.6.3 Single calls and list length

Single calls vary widely (Figure 13B). Within one arm, a call's yield ranged up to about fourfold (for example 12 to 67 top ideas per 100k tokens for Fable with the random slice), so a single brainstorming call is an unreliable sample of what a configuration delivers. In Experiment 1, with five ideas per call, about half of all calls produced no top idea at all.

Asking for 50 ideas instead of 5. For Haiku on M6, the 50-idea calls had a per-token yield close to that of the 5-idea calls with the slice (20.4 against 25.3) and higher with the other representations (for example 9.4 against 6.3 with the full graph). The two experiments differ in idea length and therefore in score, so this comparison is indicative only.

4.6.4 Style or content? A content-only re-rating

The larger models' ideas read better, and part of their higher ratings may reward presentation rather than content (Section 3.9). As an exploratory check, the 20 ideas Sonnet scored highest and the 20 it scored lowest were taken from each model (120 ideas, all on M6), shuffled together and re-rated blind by a second judge, `claude-opus-5-5` at high effort, on the same four criteria (1–10). The second judge was told to restate each idea in plain words before rating it and to disregard wording, polish, length and jargon, so that two ideas with the same mechanism would receive the same ratings. Figure 14 shows which models wrote the second judge's top 20 and bottom 20 ideas; an even split would be about 7 per model.

Table 25: Mean content rating by the second judge (Opus, 1–10, mean of four criteria) of each model's ideas, by the first judge's (Sonnet's) group. 20 ideas per cell.

Model	First judge's top 20	First judge's bottom 20
Haiku 5.5	6.20	3.33
Opus 5.5	6.75	4.31
Fable 5.1	6.71	4.20

Result. The second judge's top 20 came 3 from Haiku, 9 from Opus and 8 from Fable, and its bottom 20 15, 2 and 3. It agreed with the first judge at both ends: all of its top 20

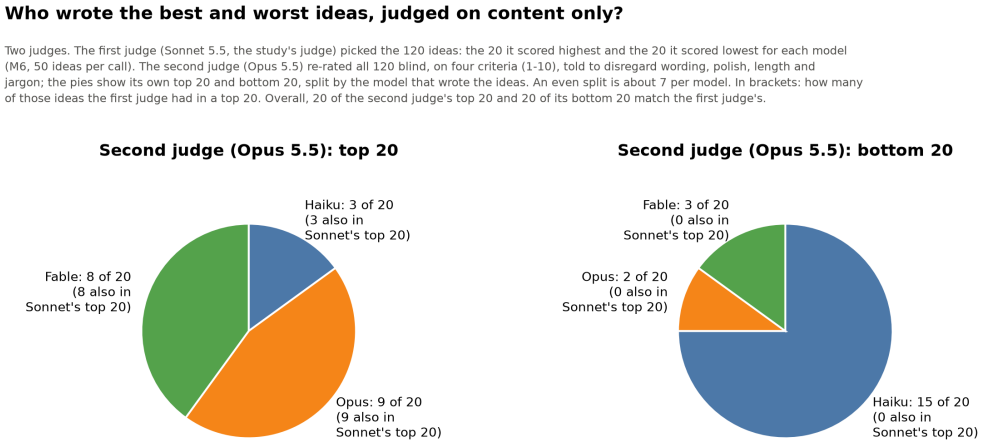


Figure 14: Content-only re-rating of the extremes. The first judge (Sonnet, the study's judge) supplied each model's 20 highest- and 20 lowest-scored ideas; the second judge (Opus) re-rated all 120 blind, told to disregard wording and polish. Left: the second judge's top 20, by the model that wrote them; right: its bottom 20. In brackets: how many of those ideas were also in the first judge's top 20 for that model.

had been in the first judge's top 20s and all of its bottom 20 in the first judge's bottom 20s (Spearman $\rho = 0.87$ over the 120 ideas). Once wording was set aside, Haiku's ideas remained the weakest at both ends, so its gap to the larger models is one of content. Opus and Fable were indistinguishable on content (mean ratings 6.75 and 6.71 for their top ideas, 4.31 and 4.20 for their bottom ideas; Table 25), so Fable's higher yield under the first judge may partly reflect presentation.

Caveats. The second judge is one of the three generators and may still favour its own style, although it saw no authorship. Forty ideas per model, chosen at the extremes of another judge's ranking, are a small and selective sample; the check bounds the role of presentation rather than measuring it. Twelve ideas tied at the cut of the top 20 (content rating 7.0); ties were broken by the first judge's score, which slightly favours Opus and Fable, whose top scores reach 4.50 against 4.25 for Haiku. Random tie-breaking would move at most one or two places and leaves the conclusion unchanged.

4.7 Robustness and caveats

4.7.1 Length and judge reliability

Table 26 checks whether the judge simply preferred longer sets.

Table 26: Length check: mean words per set in each comparison, how often the longer set won, and the winning arm's win rate.

Comparison	Mean words per set (first / second arm)	Longer set won	Win rate of the winning arm
A · Full Graph vs B · Slice, low effort	452 / 443	65%	65%, A · Full Graph
A · Full Graph vs B · Slice, high effort	542 / 492	71%	64%, A · Full Graph

Table 26 (continued)

Comparison	Mean words per set (first / second arm)	Longer set won	Win rate of the winning arm
High vs low effort, A · Full Graph	542 / 452	90%	92%, high
High vs low effort, B · Slice	492 / 443	85%	94%, high
xhigh vs high effort, A · Full Graph	587 / 530	88%	73%, xhigh
xhigh vs high effort, B · Slice	587 / 499	83%	85%, xhigh
B · Slice · high vs A · Full Graph · low	492 / 452	74%	91%, B · Slice · high
A · Full Graph vs C · Control, high effort	542 / 529	68%	74%, A · Full Graph
B · Slice vs C · Control, high effort	492 / 529	60%	62%, B · Slice
B · Slice vs R · Random, high effort	492 / 492	59%	54%, B · Slice

- **A vs B at low effort:** sets are nearly the same length, so A's set-level advantage is not explained by length.
- **Effort:** each step up produces longer sets, so part of the effort effect may be detail bias in the judge.
- **B-high vs A-low:** B-high wins 91% but the longer set wins only 74%, so B-high also wins pairs where its ideas are shorter.
- **B vs C:** B wins although its sets are shorter, so the graph's advantage over no graph is not a length effect.
- **Order agreement:** 68–97% of pairs got the same verdict in both orders (85% overall); split verdicts count as ties. The lowest is B vs R (68%), where the sets are hardest to tell apart.
- **Individual scores** do not depend on length comparisons between arms, which makes the tail measure less exposed to detail bias than the pairwise judging.

4.7.2 Change from the pilot

The pilot (2 market needs × 3 replicates) found the full graph winning 17 of 18 set comparisons. At full scale (13 market needs, 65 pairs) A's set-level advantage shrinks to about 64%, and on the tail measure it disappears. The pilot's "more novel but less feasible" pattern for A did not replicate (it now describes no graph, C). The effort effect and the B-high vs A-low result held. Contrary to the pilot's expectation, quality kept rising from high to xhigh.

4.7.3 Caveats

- **Single judge model,** with no human calibration yet; a "top idea" is one the judge rates highly on relevance, novelty, feasibility and specificity together, a grounded strong idea rather than necessarily the wildest one.
- **Coarse scores:** individual scores move in 0.25 steps, so the threshold (4.0, top 12.3%) is the nearest available cut, and the 4.25 tail has too few ideas to separate arms.
- **Yield assumes a constant rate per token.** The observed running counts are close to straight lines (Figure 7), but they depend on the order in which calls are taken; the slope itself does not.
- **Budget curves assume independent calls** within a market need and use a small-sample estimate of each market need's success rate; they are a model, not observed sequences.

- **R and C were run only at high effort;** R vs C was not judged pairwise, so whether a random slice beats no graph on whole sets is unknown.
- **Relevance vs size rests on the tail measure:** the pairwise judging found no difference between B and R.
- **R controls for node count, not edges or graph distribution:** R matches B in node count and layer composition, but random selection yields substantially sparser induced graphs (2–52 edges vs. 11–85 in B) and 17% fewer input tokens. Furthermore, only a single fixed random draw per market need was evaluated rather than an ensemble over multiple random graph realizations, so variance across different random subgraphs is not averaged out.
- **Length and effort are intertwined;** the pairwise effort effects may be partly inflated by detail bias.
- **xhigh covers 6 of 13 market needs,** which are slightly easier than average for top ideas; see the matched-needs results.
- **One product, one operator, one prompt.** The worked example comes from the M8 branch of the graph, which may favour ideas in that area.
- **Per-market-need results** rest on 5 calls or pairs each and should be read as patterns; the slice-size effect is exploratory (13 market needs, $p \approx 0.04$–0.13).

5 Discussion

This study asked how much of a product's knowledge graph an LLM needs to see in order to brainstorm well, judged not by the quality of its output alone but by how efficiently it turns tokens, and money, into top ideas. An innovation matrix of a microwave oven (123 nodes, 246 edges) was given to a model in four forms: the full graph (A), the vertical slice for the target market need (B), a random slice of the same size (R), or no graph at all (C). In Experiment 1, `claude-haiku-5-5` applied the SUBSTITUTE operator to each of 13 market needs at four levels of thinking effort, producing 2,892 ideas. In Experiment 2, Haiku, `claude-opus-5-5` and `claude-fable-5-1` were each asked for 50 ideas per call on one market need. All ideas were judged blind by `claude-sonnet-5-5`. Brainstorming is a search for the rare great idea, so the main measure was the **yield**: top ideas (score ≥ 4.0) per 100k tokens, and in Experiment 2 also per dollar.

The short answer is that, for finding top ideas, the knowledge graph did not pay for itself. Six conclusions support it.

1. **The full graph never paid off.** In Experiment 1 it yielded about 3 top ideas per 100k tokens at every effort, against 8.8 for the market-need slice at medium effort and 6.4 for no graph; the slice was more efficient on 10 of 13 market needs (sign test $p = 0.039$). In Experiment 2 it was the least efficient representation for all three models, at about half the yield per token of the slice or no graph for Haiku and Fable.

2. **A relevant slice removes the penalty but adds no gain.** If a graph is given, the market-need slice is the most efficient form, and its lead over the full graph looks largest where the slice is small (exploratory). It is not clearly better than giving no graph: in Experiment 1 no graph was run at high effort only, and in Experiment 2, with every representation at the same effort, the slice and no graph were indistinguishable for Haiku, Opus and Fable alike.

3. **More context buys better average sets, not more great ideas.** Judged as whole sets, the full graph beats the slice (64%) and no graph (74%), mainly by being more specific. But its top-idea rate is no higher than the slice's, and its calls cost about twice as much. For Opus and Fable the full graph did not lower the share of top ideas at all; its whole penalty was the cost of the extra input.

4. **Irrelevant context hurts the small model most.** For Haiku, a random slice of the same size as the market-need slice is less efficient than both the slice and no graph (5.9 and 6.4 against 4.1 at high effort; no graph beats it on 10 of 12 market needs, $p = 0.039$), although whole-set judging cannot tell the two slices apart. For Opus a random slice cost nothing against no graph, and for Fable less than the full graph did.

5. **Thinking effort raises quality, not efficiency.** High effort beats low in about 93% of set comparisons, and extra-high effort gives the highest top-idea rates (19–21%). But each call costs proportionally more, so yield does not rise with effort: the full graph stays at about 3 top ideas per 100k tokens from low to extra-high, and the slice peaks at medium.

6. **Model choice is a question of unit economics.** Per token, Opus and Fable found roughly 1.5 to 2.5 times as many top ideas as Haiku; per dollar at list prices, Haiku found about 20 times more than Opus and 40 times more than Fable. The best model per token is the worst per dollar (exploratory; Section 4.6).

These results are bounded. They concern one product, one operator, one judge model and, in Experiment 2, one market need, and a top idea is one this judge rated highly, not an idea checked against prior art. The expert reading suggests what the judge rewards: Haiku behaves as a retrieval system, holding a lossy copy of its training corpus, patents and product literature included, and surfacing the engineering moves that bear on a stated problem (Section 3.10). A retrieval system is most efficient when the query is precise, and a stated market need is already a precise query; a large graph adds cost more than direction. Whether larger models do more than retrieve is open. It has been argued that language models are structurally incapable of the abductive "jump" to new premises (Zahavy, 2026), and that fixed-weight transformer inference lacks the coupling between error and cost that such a jump requires, regardless of model scale (Balani and Panda, 2026); Experiment 2 is consistent with that view but cannot test it. A content-only re-rating (Section 4.6.4) suggests that their lead over Haiku is one of content, while Fable's edge over Opus may be one of presentation; checking the best ideas against prior art would show whether either goes beyond retrieval.

For practice, the advice is plain: state the market need, give the model at most the slice of the graph that serves it, and choose the model by cost per top idea rather than by its reputation.

Future work. Four extensions follow from the limits of this study. First, the many settings that change cost and yield (representation, model, effort, persona, list length) suggest treating prompt configuration as an optimisation problem: Bayesian optimisation or bandit methods could search this space for the cheapest setting that reaches a target number of top ideas for a given product and need, rather than testing a few settings exhaustively. Second, the judge's ratings should be anchored outside the judge: rewriting all ideas into one neutral style before judging would separate content from presentation, and checking the best ideas against patents and products would show whether any of them goes beyond the state of the art. Third, the move coding of Experiment 2, planned before it was run, will show whether the representation changes how many distinct ideas a model reaches, not only how many it rates highly. Finally, the results rest on one product, one operator and one judge; repeating them across products, operators such as TRIZ, and human raters would show how far they generalise.

5.1 Implications for design thinking

For design practice, the results point the same way as a familiar design move: framing beats volume. Stating the market need precisely did most of the work. Giving the model the relevant slice of the product graph was the machine version of narrowing the problem, but it did no better than the stated need alone, and the full graph only added cost. Three highlights:

The measure introduced here, *yield* (top ideas per token or per dollar), turns ideation into a budgetable activity: the cost of a given number of strong candidates can be estimated before

a session is run, which may also help to design and tune agentic ideation workflows.

None of this makes the product graph useless. It remains a strong tool for systematic search, for example a mutually exclusive, collectively exhaustive (MECE) decomposition of the design space, which could help make the combinatorics of innovation explicit rather than leaving them to the model's recall.

Finally, we propose, and leave for future work, the idea of a *diffable* product graph: one that can be compared with another graph to show exactly what differs. Diffing a product's graph against a competitor's, against an earlier version, or against the graph of the needs it should serve would point to where innovation is missing or possible, giving agents a machine friendly operator to find explore the design space.

References

Yoji Akao, editor. *Quality Function Deployment: Integrating Customer Requirements into Product Design*. Productivity Press, New York, 2024. doi: 10.4324/9781003578833. Reissue of the 1990 edition.

Chenxin An, Jun Zhang, Ming Zhong, Lei Li, Shansan Gong, Yao Luo, Jingjing Xu, and Lingpeng Kong. Why Does the Effective Context Length of LLMs Fall Short? arXiv preprint arXiv:2410.18745, 2024.

Barrett R Anderson, Jash Hemant Shah, and Max Kreminski. Homogenization Effects of Large Language Models on Human Creative Ideation. *Creativity and Cognition 2024*, 2024. doi: 10.1145/3635636.3656204.

Joshua Ashkinaze, Julia Mendelsohn, Li Qiwei, Ceren Budak, and Eric Gilbert. How AI Ideas Affect the Creativity, Diversity, and Evolution of Human Ideas: Evidence From a Large, Dynamic Experiment. *Proceedings of the ACM Collective Intelligence Conference 2025*, 2025. doi: 10.1145/3715928.3737481.

Paras Balani and Subhrakanta Panda. LLMs don't pay for the jump, 2026.

Jose Berengueres. *The Brown Book of Design Thinking*. UAE University College, Al Ain, 2023.

T. R. Browning. Applying the design structure matrix to system decomposition and integration problems: a review and new directions. *IEEE Transactions on Engineering Management 48(3)*, 2001. doi: 10.1109/17.946528.

Lai-Kow Chan and Ming-Lu Wu. Quality function deployment: A literature review. *European Journal of Operational Research 143(3)*, 2002. doi: 10.1016/s0377-2217(02)00178-9.

Lingjiao Chen, Matei Zaharia, and James Zou. FrugalGPT: How to Use Large Language Models While Reducing Cost and Improving Performance. *Transactions on Machine Learning Research*, 2024a.

Liuqing Chen, Qianzhi Jing, Yixin Tsang, Qianyi Wang, Lingyun Sun, and Jianxi Luo. DesignFusion: Integrating Generative Models for Conceptual Design Enrichment. *Journal of Mechanical Design*, 2024b. doi: 10.1115/1.4065487.

Liuqing Chen, Yaxuan Song, Shixian Ding, Lingyun Sun, Peter Childs, and Haoyu Zuo. TRIZ-GPT: An LLM-Augmented Method For Problem-Solving. *ASME IDETC-CIE 2024*, 2024c. doi: 10.1115/detc2024-143163.

Liuqing Chen, Yiyan Tsang, Qianzhi Jing, and Lingyun Sun. A LLM-augmented Morphological Analysis Approach for Conceptual Design. *Proceedings of DRS2024*, 2024d. doi: 10.21606/drs.2024.605.

Liuqing Chen, Haoyu Zuo, Zebin Cai, Yuan Yin, Yuan Zhang, Lingyun Sun, Peter Childs, and Boheng Wang. Toward Controllable Generative Design: A Conceptual Design Generation Approach Leveraging the Function–Behavior–Structure Ontology and Large Language Models. *Journal of Mechanical Design*, 2024e. doi: 10.1115/1.4065562.

Ted Chiang. ChatGPT Is a Blurry JPEG of the Web. *The New Yorker*, February 2023. URL `https://www.newyorker.com/tech/annals-of-technology/chatgpt-is-a-blurry-jpeg-of-the-web`.

Ely Dahan and Haim Mendelson. An Extreme-Value Model of Concept Testing. *Management Science*, 47(1):102–116, 2001. doi: 10.1287/mnsc.47.1.102.10666.

Dujian Ding, Ankur Mallick, Chi Wang, et al. Hybrid LLM: Cost-Efficient and Quality-Aware Query Routing. arXiv preprint arXiv:2404.14618, 2024.

Shiying Ding, Xinyi Chen, Yan Fang, Wenrui Liu, Yiwu Qiu, and Chunlei Chai. DesignGPT: Multi-Agent Collaboration in Design. *2023 16th International Symposium on Computational Intelligence and Design (ISCID)*, 2023. doi: 10.1109/iscid59865.2023.00056.

Anil R. Doshi and Oliver P. Hauser. Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science Advances*, 2024. doi: 10.1126/sciadv.adn5290.

Bob Eberle. *SCAMPER: Games for Imagination Development*. D.O.K. Publishers, Buffalo, NY, 1971.

Steven D. Eppinger and Tyson R. Browning. *Design Structure Matrix Methods and Applications*. MIT Press, Cambridge, MA, 2012. doi: 10.7551/mitpress/8896.001.0001.

Lee Fleming. Recombinant Uncertainty in Technological Search. *Management Science 47(1)*, 2001. doi: 10.1287/mnsc.47.1.117.10671.

J. S. Gero. Design Prototypes: A Knowledge Representation Schema for Design. *AI Magazine 11(4)*, 1990. doi: 10.1609/aimag.v11i4.854.

John S. Gero and Udo Kannengiesser. The situated function–behaviour–structure framework. *Design Studies 25(4)*, 2004. doi: 10.1016/j.destud.2003.10.010.

Karan Girotra, Christian Terwiesch, and Karl T. Ulrich. Idea Generation and the Quality of the Best Idea. *Management Science*, 56(4):591–605, 2010. doi: 10.1287/mnsc.1090.1144.

Tingxu Han, Zhenting Wang, Chunrong Fang, et al. Token-Budget-Aware LLM Reasoning. *Findings of the Association for Computational Linguistics: ACL 2025*, 2025. doi: 10.18653/v1/2025.findings-acl.1274.

John R. Hauser and Don Clausing. The House of Quality. *Harvard Business Review*, 66(3):63–73, 1988.

Julie Hirtz, Robert B. Stone, Daniel A. McAdams, Simon Szykman, and Kristin L. Wood. A functional basis for engineering design: Reconciling and evolving previous efforts. *Research in Engineering Design 13(2)*, 2002. doi: 10.1007/s00163-001-0008-3.

Shuo Jiang, Weifeng Li, Yuping Qian, Yangjun Zhang, and Jianxi Luo. AutoTRIZ: Automating engineering innovation with TRIZ and large language models. *Advanced Engineering Informatics*, 2025. doi: 10.1016/j.aei.2025.103312.

Sayash Kapoor, Benedikt Stroebl, Zachary S. Siegel, et al. AI Agents That Matter. *Transactions on Machine Learning Research*, 2025.

Andrej Karpathy. Intro to Large Language Models. Video Lecture, 2023. URL `https://www.youtube.com/watch?v=zjkBMFhNj_g`.

Steven Kyffin and Paul Gardien. Navigating the Innovation Matrix: An Approach to Design-led Innovation. *International Journal of Design*, 3(1):57–69, 2009.

Mosh Levy, Alon Jacoby, and Yoav Goldberg. Same Task, More Tokens: the Impact of Input Length on the Reasoning Performance of Large Language Models. *Proceedings of ACL 2024 (Long Papers)*, 2024. doi: 10.18653/v1/2024.acl-long.818.

Zhuowan Li, Cheng Li, Mingyang Zhang, et al. Retrieval Augmented Generation or Long-Context LLMs? A Comprehensive Study and Hybrid Approach. *Proceedings of EMNLP 2024: Industry Track*, 2024. doi: 10.18653/v1/2024.emnlp-industry.66.

Nelson F. Liu, Kevin Lin, John Hewitt, et al. Lost in the Middle: How Language Models Use Long Contexts. *Transactions of the Association for Computational Linguistics*, 2024. doi: 10.1162/tacl_a_00638.

Songkai Liu, Yanqing Shen, Yilun Zhang, Zhangli Hou, Xin Wang, Jianxi Luo, and Zhinan Zhang. iDesignGPT enhances conceptual design via large language model agentic workflows. *Nature Communications*, 2026. doi: 10.1038/s41467-026-68672-1.

Jianxi Luo, Serhad Sarica, and Kristin L. Wood. Guiding data-driven design ideation by knowledge distance. *Knowledge-Based Systems 218*, 2021. doi: 10.1016/j.knosys.2021.106873.

Kevin Ma, Daniele Grandi, Christopher McComb, and Kosa Goucher-Lambert. Conceptual Design Generation Using Large Language Models. *ASME IDETC-CIE 2023*, 2023. doi: 10.1115/detc2023-116838.

Kevin Ma, Daniele Grandi, Christopher McComb, and Kosa Goucher-Lambert. Do Large Language Models Produce Diverse Design Concepts? A Comparative Study with Human-Crowdsourced Solutions. *Journal of Computing and Information Science in Engineering*, 2025. doi: 10.1115/1.4067332.

Soheyl Massoudi and Mark Fuge. Agentic Large Language Models for Conceptual Systems Engineering and Design. *Journal of Mechanical Design*, 2026. doi: 10.1115/1.4070328.

Ali Modarressi, Hanieh Deilamsalehy, Franck Dernoncourt, et al. NoLiMa: Long-Context Evaluation Beyond Literal Matching. arXiv preprint arXiv:2502.05167, 2025.

Isaac Ong, Amjad Almahairi, Vincent Wu, et al. RouteLLM: Learning to Route LLMs with Preference Data. arXiv preprint arXiv:2406.18665, 2024.

Alex F. Osborn. *Applied Imagination: Principles and Procedures of Creative Problem-Solving.* Charles Scribner's Sons, New York, 1953.

Norman Paulsen. Context Is What You Need: The Maximum Effective Context Window for Real World Limits of LLMs. *Advances in Artificial Intelligence and Machine Learning*, 2026. doi: 10.54364/AAIML.2026.61268.

Robert Phaal, Clare J. P. Farrukh, and David R. Probert. Technology roadmapping—A planning framework for evolution and revolution. *Technological Forecasting and Social Change 71(1–2)*, 2004. doi: 10.1016/s0040-1625(03)00072-6.

T. Ritchey. Problem structuring using computer-aided morphological analysis. *Journal of the Operational Research Society 57(7)*, 2006. doi: 10.1057/palgrave.jors.2602177.

Tom Ritchey. General morphological analysis as a basic scientific modelling method. *Technological Forecasting and Social Change 126*, 2018. doi: 10.1016/j.techfore.2017.05.027.

Serhad Sarica, Jianxi Luo, and Kristin L. Wood. TechNet: Technology semantic network based on patent data. *Expert Systems with Applications 142*, 2020. doi: 10.1016/j.eswa.2019.112995.

Prabir Sarkar and Amaresh Chakrabarti. Assessing design creativity. *Design Studies 32(4)*, 2011. doi: 10.1016/j.destud.2011.01.002.

Jami J. Shah, Steve M. Smith, and Noe Vargas-Hernandez. Metrics for measuring ideation effectiveness. *Design Studies*, 2003. doi: 10.1016/s0142-694x(02)00034-0.

Chenglei Si, Diyi Yang, and Tatsunori Hashimoto. Can LLMs Generate Novel Research Ideas? A Large-Scale Human Study with 100+ NLP Researchers. arXiv preprint arXiv:2409.04109, 2024.

Donald V. Steward. The design structure system: A method for managing the design of complex systems. *IEEE Transactions on Engineering Management EM-28(3)*, 1981. doi: 10.1109/tem.1981.6448589.

Robert B. Stone and Kristin L. Wood. Development of a Functional Basis for Design. *Journal of Mechanical Design 122(4)*, 2000. doi: 10.1115/1.1289637.

Kamil Szczepanik and Jarosław Chudziak. TRIZ Agents: A Multi-Agent LLM Approach for TRIZ-Based Innovation. *Proceedings of ICAART 2025 (SciTePress)*, 2025. doi: 10.5220/0013321900003890.

Christian Terwiesch, Lennart Meincke, Karan Girotra, Ethan Mollick, Gideon Nave, and Karl Ulrich. Artificial intelligence and its impact on creativity and diversity: An empirical study of large language model-generated product ideas. *Production and Operations Management*, 2026. doi: 10.1177/10591478261474243.

Brian Uzzi, Satyam Mukherjee, Michael Stringer, and Ben Jones. Atypical Combinations and Scientific Impact. *Science 342(6157)*, 2013. doi: 10.1126/science.1240474.

Dennis Verhoeven, Jurriën Bakker, and Reinhilde Veugelers. Measuring technological novelty with patent-based indicators. *Research Policy 45(3)*, 2016. doi: 10.1016/j.respol.2015.11.010.

Samangi Wadinambiarachchi, Ryan M. Kelly, Saumya Pareek, Qiushi Zhou, and Eduardo Velloso. The Effects of Generative AI on Design Fixation and Divergent Thinking. *Proceedings of CHI 2024*, 2024. doi: 10.1145/3613904.3642919.

Junlin Wang, Siddhartha Jain, Dejiao Zhang, et al. Reasoning in Token Economies: Budget-Aware Evaluation of LLM Reasoning Strategies. *Proceedings of EMNLP 2024*, 2024. doi: 10.18653/v1/2024.emnlp-main.1112.

Ningning Wang, Xavier Hu, Pai Liu, et al. Efficient Agents: Building Effective Agents While Reducing Cost. arXiv preprint arXiv:2508.02694, 2025.

Tom Zahavy. Position: LLMs Can't Jump. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2026. URL `https://icml.cc/virtual/2026/poster/67091`.

Qihao Zhu and Jianxi Luo. Generative Transformers for Design Concept Generation. *Journal of Computing and Information Science in Engineering*, 2022. doi: 10.1115/1.4056220.

F. Zwicky. The Morphological Approach to Discovery, Invention, Research and Construction. In *New Methods of Thought and Procedure*, pages 273–297. Springer, 1967. doi: 10.1007/978-3-642-87617-2_14.

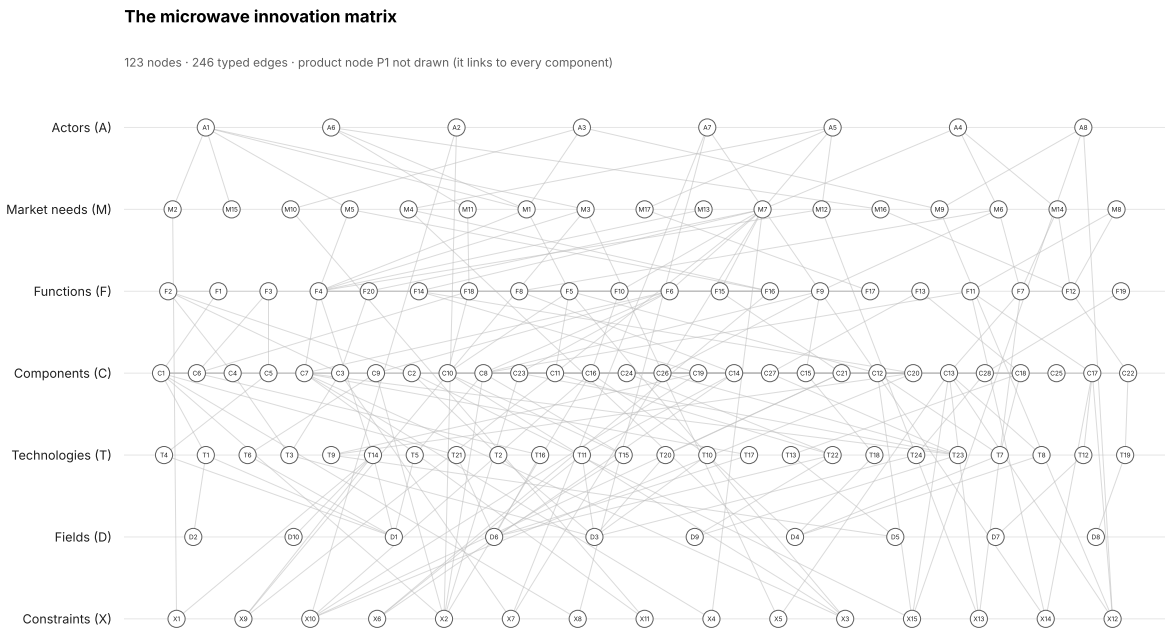


Figure 15: The full microwave innovation matrix (condition A).

A The microwave innovation matrix

Figure 15 shows the whole graph. All 123 nodes are listed below by layer; the 246 edges and their types are in `microwave_base_graph.yaml`.

Market needs (17)

ID	Label
M1	Fast reheating for busy households
M2	Crispy, browned texture like conventional oven
M3	Hot milk / drink ready for breakfast
M4	Defrost frozen food quickly
M5	Save time in the kitchen
M6	Simple operation with no learning curve
M7	Safe for family use (radiation, burns, electric shock)
M8	Human need for feedback (know when food is ready)
M9	Low purchase price
M10	Small kitchen footprint
M11	Easy to clean
M12	Healthy diet (nutrient retention, less fat)
M13	Energy efficiency
M14	Accessible for elderly / visually impaired users
M15	Appliances that make you feel better (latent need)
M16	Need for mobility (not having to stand by the appliance)
M17	Bake a birthday cake (prep tasks like melting butter)

Functions (20)

ID	Label
F1	Generates microwave energy at 2.45 GHz
F2	Converts mains voltage to ∼4 kV DC for the magnetron
F3	Guides microwave energy into the cavity
F4	Heats food via dielectric water-molecule vibration (FastHeating)

ID	Label
F5	Distributes heat evenly by rotating food through the field
F6	Contains microwave radiation inside the appliance
F7	Sets and counts down cooking duration
F8	Sets average heating power level
F9	Opens and closes the door to access the cavity
F10	Cuts RF power instantly when the door opens
F11	Illuminates cavity so the user can watch the food
F12	Signals end of cooking (audible alert)
F13	Cools magnetron and electronics
F14	Protects against overheating
F15	Protects against overcurrent and electrical faults
F16	Defrosts frozen food (low-power cycling)
F17	Melts / softens butter, chocolate, etc.
F18	Contains spills and allows cleaning
F19	Suppresses electrical noise conducted back to mains
F20	Discharges stored high voltage after switch-off

Components (28)

ID	Label
C1	Magnetron tube (2.45 GHz, ∼700 W output)
C2	High-voltage transformer
C3	High-voltage capacitor (∼1 μF, 2.1 kV) with internal bleeder resistor
C4	High-voltage rectifier diode
C5	Waveguide duct
C6	Mica waveguide cover
C7	Enamelled steel cooking cavity (∼20 L)
C8	Hinged door with perforated metal mesh window
C9	Door choke seal (quarter-wave rim)
C10	Rotating glass turntable tray
C11	Roller ring and coupler (turntable support)
C12	Turntable motor (∼5 rpm)
C13	Mechanical timer knob & cam switch (0-30 min)
C14	Power-level knob (5-step duty-cycle cam switch)
C15	Door-release push button & latch lever
C16	Door interlock microswitches (primary, secondary, monitor)
C17	Interior lamp (20-25 W incandescent)
C18	Cooling fan and fan motor
C19	Magnetron thermal cutout
C20	Cavity thermal cutout
C21	Main ceramic fuse
C22	End-of-cycle bell (mechanical ‘ding’)
C23	Outer steel cabinet (wrapper)
C24	Control panel fascia
C25	Mains power cord and plug
C26	Door hinges
C27	Ventilation louvres
C28	Mains EMI noise filter

Technologies (24)

ID	Label
T1	Electron cyclotron motion in crossed $E \times B$ fields (cavity magnetron)
T2	Magnetic-shunt leakage HV transformer
T3	Half-wave voltage doubler (capacitor + diode)
T4	Rectangular waveguide TE10 mode propagation

ID	Label
T5	Sub-wavelength aperture shielding (Faraday cage effect)
T6	Quarter-wave choke seal
T7	Shaded-pole / synchronous AC micro-motor (electromagnetic induction)
T8	Motor-driven cam timer with spring-return mechanism
T9	Bimetallic snap-disc thermostat
T10	Redundant interlock microswitch logic (primary / secondary / monitor)
T11	Duty-cycle power modulation (on/off cycling)
T12	Incandescent tungsten-filament lamp
T13	Forced-air convection cooling
T14	Dielectric heating (dipolar rotation of water molecules)
T15	Tempered / borosilicate glass forming
T16	Mica sheet dielectric (muscovite laminate)
T17	Enamel / epoxy powder coating of steel
T18	Spring-loaded lever latch mechanics
T19	Mechanical bell striker
T20	Ceramic cartridge fuse (fusible-link overcurrent protection)
T21	Sintered ferrite permanent magnets
T22	Injection-molded engineering thermoplastics (PP, PBT, PPS)
T23	Sheet-steel stamping & spot welding
T24	Passive LC EMI line filtering

Fields (10)

ID	Label
D1	Radio-frequency & microwave engineering
D2	Vacuum electronics
D3	Power electronics & electrical engineering
D4	Electromechanics & mechanisms
D5	Thermal engineering & heat transfer
D6	Materials science (metals, ceramics, glass, polymers)
D7	Lighting & optics
D8	Sound engineering
D9	Manufacturing engineering (stamping, molding, assembly)
D10	Food science (dielectric properties of food)

Constraints (15)

ID	Label
X1	Microwaves can't brown or crisp food surfaces
X2	FCC/FDA radiation-leakage safety limits (21 CFR 1030.10, 5 mW/cm^2 at 5 cm)
X3	Uneven heating — standing-wave hot and cold spots
X4	Metal objects in cavity cause arcing
X5	Mechanical timer imprecise for short (<1 min) durations
X6	Power only adjustable by on/off cycling, not continuously
X7	HV capacitor can hold a lethal charge after unplugging
X8	Limited magnetron lifetime (cathode/filament wear)
X9	Superheating of liquids — risk of eruptive boiling
X10	Electrical safety standards (IEC 60335-2-25 / UL 923)
X11	Cavity volume limited by countertop size
X12	Low BOM target for entry-level price segment
X13	No food-state sensing — open-loop cooking
X14	Interior lamp burns out and is hard to replace
X15	Regional mains variants (230 V/50 Hz vs 120 V/60 Hz)

Actors (8)

ID	Label
A1	Time-pressed home cook
A2	Appliance safety regulator/inspector
A3	Student / single-person household
A4	Elderly user
A5	Parent with young children
A6	Office worker using a shared kitchen
A7	Repair technician
A8	OEM manufacturer / cost engineer

Product (1)

ID	Label
P1	Basic countertop microwave oven (timer knob, power knob, door button, lamp)

B Generation prompt

The template used for every call (`template_prompt.txt`). Placeholders in braces are filled per call; the opening sentence of the system prompt is shown filled in.

```
=== SYSTEM ===
You are an innovation engineer. You read product graphs and propose radical
product innovations. Respond with valid JSON only, no prose outside the JSON.

=== USER ===
Innovate the following {product_name} based on the graph given.
Encourage radical and wild ideas.

TARGET MARKET NEED
{target_need_id}: {target_need_label}
Favour innovations that serve this market need.

INNOVATION OPERATOR: {operator_name}
{operator_definition}

EXAMPLE OF AN INNOVATION WE LIKE
A microwave has a door with holes, a light, and other components and
functions, so the user can see inside: they need to check for mistakes
(perhaps there is also entertainment value).
Applying the operator: replace the light and the holes with something
that fulfils the same function equally well or better, or more cheaply.
  - e.g. 1: a webcam
  - e.g. 2: a fully transparent microwave casing that needs no light bulb inside

GRAPH OF THE {product_name}
{graph_yaml}

OUTPUT
Return exactly {n_ideas} ideas as JSON:
{
  "ideas": [
    {
      "title": "short name",
      "anchors": ["existing node ids the operator was applied to"],
      "new_nodes": [
        {"type": "Field|Technology|Component|Function|MarketNeed|Constraint",
```

```
        "label": "...", "detail": "one line"}
      ],
      "new_edges": [
        {"from": "node id or new node label", "to": "node id or new node label",
         "type": "one of the graph's edge_types, e.g. resolves"}
      ],
      "rationale": "one sentence: how this serves {target_need_id}"
    }
  ]
}
```

C Pairwise judge prompt

System prompt:

```
You are an expert panel judging product-innovation brainstorming. You compare two
↪  sets of ideas produced for the same brief. Judge only the ideas themselves;
↪  ignore length and formatting. Respond with valid JSON only.
```

User prompt:

```
BRIEF
Product: {product}
Target market need: {need_id}: {need_label}
Innovation operator: {operator}
The brief asked for radical, wild ideas that serve the target need.

SET 1
{set1}

SET 2
{set2}

Score each set from 1 (poor) to 5 (excellent) on:
- relevance: how well the ideas serve the target need
- novelty: how radical and non-obvious the ideas are
- feasibility: how technically plausible the ideas are
- specificity: how concrete the ideas are (named components, mechanisms)
- diversity: how different the ideas are from each other

Then pick the better set overall for this brief.

Return JSON:
{"set_1": {"relevance": 0, "novelty": 0, "feasibility": 0, "specificity": 0,
↪  "diversity": 0},
 "set_2": {"relevance": 0, "novelty": 0, "feasibility": 0, "specificity": 0,
 ↪  "diversity": 0},
 "winner": "set_1" | "set_2" | "tie",
 "reason": "one sentence"}
```

D Individual scoring prompts

Quality, used for every idea. System prompt:

```
You are an expert panel judging product-innovation ideas. Rate each idea on its own
↪  merits, independently of the others. Ignore writing style and length. Respond
↪  with valid JSON only.
```

User prompt:

```
BRIEF
Product: {product}
Target market need: {need_id}: {need_label}
Innovation operator: {operator}
The brief asked for radical, wild ideas that serve the target need.

IDEAS
{ideas}

Rate EACH idea from 1 (poor) to 5 (excellent) on:
- relevance: how well it serves the target need
- novelty: how radical and non-obvious it is
- feasibility: how technically plausible it is
- specificity: how concrete it is (named components, mechanisms)

Return JSON with one entry per idea, in the same order:
{"ratings": [{"idea": 1, "relevance": 0, "novelty": 0, "feasibility": 0,
↪  "specificity": 0,
              "comment": "one sentence"}]}
```

E Data and code

The project repository (`https://github.com/orioli/IM-LLM`) is organized as follows:

Component / Path	Contents & Role
`microwave_base_graph.yaml`	Full innovation matrix (123 nodes, 246 typed edges)
`slicer.py`, `slices/`	Graph slicer generating vertical slices (B) and random slices (R) with `manifest.csv`
`KG_informed_brainstorm_AB_Test.py`	Ideation execution harness across conditions and effort levels
`analyze.py`, `top_ideas.py`	Pairwise set judging and individual quality scoring harnesses
`runs/rq1_full.jsonl`	Complete raw call logs (580 runs, 2,892 ideas, prompt hashes, token counts)
`results/judgments.jsonl`, `results/idea_scores.jsonl`	Raw blind judge evaluations and point scores
`results/pairwise_summary.csv`, `results/tail_summary.csv`	Summary statistics, win rates, and bootstrap yield slope estimates
`yield_per_market_need.xlsx`	Per-market-need top ideas, tokens, yields, and sign tests (live spreadsheet formulas)
`render_matrix_v2.py`, `*.html`	Interactive visualizers for exploring the full matrix and subgraph slices
`charts.py`, `tail_charts.py`	Figure and visualization generation scripts
`build_paper.py`	Automated end-to-end manuscript build, data verification, and compilation pipeline